Risky business: Corporate governance in Indian banks

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This version: August 25, 2020
Preliminary draft. Please do not cite or circulate.

Abstract

In this paper, I study corporate governance in Indian banks along three dimensions. First, I provide detailed description of what public and private bank boards look like, and how they differ from each other. Public banks face several major external constraints when it comes to their board members, but I focus in particular on tenure and compensation. I then investigate whether bank valuations respond significantly to appointments and resignations of chief executive officers (CEOs) and independent directors. I find that shareholders consider turnover of CEOs to be positive news in public banks, but not in private banks. For public banks, appointments of independent directors are also considered positively; their resignations negatively. Gender of the CEO plays an important role. Finally, I analyse how bank risk is associated with the CEO's age, tenure, and compensation. I find that increasing CEO tenure and age, and decreasing the variable component of pay for CEOs is in particular linked to lower bank risk in public banks.

JEL classification: E58, G21, G34, G38
Keywords: Corporate governance; board of directors; banking regulation; executive compensation; Indian banking sector.

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Work on this paper was done as part of my PhD thesis at The Graduate Institute of International and Development Studies (IHEID), before joining the BoE.
I would especially like to thank Ugo Panizza for his comments and help in obtaining the data; and Pramod Sinha at National Institute of Public Finance & Policy (NIPFP) for help in accessing and understanding Indian bank balance sheet data. I am also grateful to Ryan Banerjee, Shekhar Hari Kumar, Rahul Mukherjee, Steven Ongena, Rajeswari Sengupta, Ajay Shah, and Cédric Tille for their comments at various stages of the paper. I benefited from comments by seminar participants at the *Brown Bag Lunch* at IHEID, and *Econalanya* 2018. The views expressed in this paper are my own, and not necessarily those of the Bank of England. All errors are my own.

1 Introduction

Corporate governance issues in banks are critical and likely to be different from those at non-financial firms. They have also re-surfaced in a big way after various scandals exposed weaknesses in oversight and governance at the largest banks after the global financial crisis. In general, the experiences, beliefs, and backgrounds of top executives affect their choices, and through their choices, the performance and strategic direction of the organisation (Hambrick and Mason, 1984; Wang *et al.*, 2016, for example). Therefore, it is imperative to understand how senior management and the board make decisions, monitor the bank and take on risks, what constraints they face, how they are incentivised, and what the implications of those decisions are for the bank as a whole. In this paper, I study corporate governance in Indian banks and how it is linked to bank risk-taking, and whether changes in board composition matter to bank owners (shareholders).

The Indian banking sector is dominated by public banks, and they play a vital role in providing banking access and financing to underserved sectors of the Indian economy.[1] Using India as a case study allows for a deeper investigation into the unique governance challenges faced by these banks, which is also a relatively under-studied area in the literature. Additionally, since 2011, the Indian banking sector as a whole has been plagued with a low asset quality, low profitability, and high leverage, with public banks significantly underperforming private ones.[2] For example, at the end of 2018, gross non-performing assets stood at 14% for state banks, as opposed to 3% for private banks (left panel, figure 1).

At the same time, severe lapses in oversight, several conflicts of interest, and large-scale frauds have been identified in the banking sector in the past few years, specially in state banks. At the end of 2017, the banking sector reported $2.5bn in aggregate frauds, of which the public sector accounted for roughly 70% (figure 2).[3] These developments have prompted several questions regarding the effectiveness of monitoring and corporate governance at these institutions. My paper seeks to contribute to these discussions along three dimensions.

The first is a descriptive analysis of what Indian public sector bank (PSB) boards look like in comparison to private bank boards. State banks are much larger on average as compared to private ones. Accordingly, they also have larger boards on average (mandated by the acts under which they are set up), who meet much more regularly and have more sub-committees charged with specific functions. In terms of composition, they have a higher share of women and executive directors, and fewer independent directors with respect to private sector banks. The median director in a PSB is slightly younger, has a significantly shorter tenure, and is paid less than *one-third* than her private sector counterpart. She also holds more number of "outside" directorships, i.e. positions on boards (both financial and non-financial) outside the bank, usually as a government appointee in other public firms.

[1]In March 2018, public sector banks made up roughly 60% of total outstanding credit (RBI, 2018); and they have roughly 70% of market share based on total deposits.

[2]The Economic Survey of India, 2017, has referred to it as one part of the "twin balance sheet" problem.

[3]The monetary burden of banking sector frauds is likely to be even higher for 2018. In early 2018, the state-run lender *Punjab National Bank* came under scrutiny for the largest ever bank fraud in India's history, totalling over $2bn in value. See more here.

The next research question is whether shareholders consider changes in the board of directors as important "news". I focus on two specific classes of directors: the chief executive officer (CEO) and independent directors. The CEO is the head of the board of directors, and handles all strategic decisions of the bank with the aim of maximizing shareholder value. Independent directors have no direct ties with management and are experts appointed to monitor the board's functioning. Employing an event study methodology, I find that shareholders of public sector banks react significantly positively to CEO appointments and cessations.[4] Over a window of three days around the event, public bank stock prices show a cumulative abnormal return of roughly +1.7% across both types of events. Private bank shareholders respond significantly negatively on the day of the appointment announcement to the tune of −2%, but this peters out over the next two days. My results are qualitatively similar to what Pessarossi and Weill (2013) observe for Chinese banks.

Next, I study the main drivers of abnormal returns associated with CEO turnover in public sector banks. I find that balance sheet characteristics matter more on average than director specific factors. For example, stock prices respond by less to CEO turnover in banks with healthier balance sheets i.e. higher profitability and lower net non performing assets, but they are higher if the CEO is female, younger, or more educated. These results are different from Nguyen *et al.* (2015), who find for US banks that age and education are significantly positively related to abnormal returns around appointments, whilst gender is not.

I find that independent director appointments for Indian public sector banks are shareholder value creating: the cumulative abnormal stock market return increases significantly by +1% after two days. On the other hand, cessations are associated with negative returns. There is therefore an important role for independent directors as effective monitors. This result is also qualitatively similar to Nguyen and Nielsen (2010) who show that sudden deaths of independent directors for US firms are associated with an average stock price decline of 0.85%.

Finally, I turn to the question of how bank risk is associated with board characteristics, specifically on tenure, age, and compensation. In doing so, I rely on differences in regulation between public and private sector banks in India. Public banks in India are subject to dual regulation, both by the government as well as the central bank; while private banks are only subject to regulation by the central bank. A number of committee reports on corporate governance in India in recent years have identified this dual regulation as a major hurdle that prevents a level playing field between private and public sector banks (Nayak Committee Report, 2014).[5] For example, all directors in state banks are subject to a retirement age of 60 years, and the preference given to seniority implies that the median director who gets hired is close to 58 years old. Her tenure is therefore much shorter.

My primary measure for risk is the z-score. It remained stable between 2006-2013, and

[4] In the rest of the paper, I use "cessation" to imply any reason why a director may leave a board. More information on reasons why directors leave their positions is in section 4.

[5] Cole (2009) finds that government owned bank lending tracks the electoral cycle, while the same is not true for non-election years or for private banks. He argues that public banks in India do not face hard budget constraints and are subject to political regulation. They remain corporate entities but their board of directors are made up of political appointees. There is also substantial means for the ruling political party, specially at the state level, to influence banks.

has declined since then (figure 9), indicating that the banking sector has become riskier. In a standard panel setup, I find that increasing CEO tenure, specially relative to the board, is associated with lesser riskiness for public banks and higher riskiness for private banks. CEO age is in general associated with safer banks, which is consistent with age being a proxy for past experience. On the other hand, a higher share of salary (or cash) component and lower variable component of CEO pay are linked to lower riskiness for public banks. This is primarily because the design of variable pay in public banks provides the CEO with incentive to expand aggressively and take on higher risk during her short tenure (which is capped by retirement).

In recent years, there has been a massive interest in corporate governance of state banks in India. In 2015, a Bank Boards Bureau (BBB) was created under the aegis of the *Indradhanush plan 2015* to make recommendations to the Ministry of Finance for selections and appointments of new managing directors, CEOs, and non-executive chairpersons of public sector banks.[6] Echoing several other committee reports that came before it (Nayak Committee Report, 2014, for example), the BBB has also made recommendations to overhaul the corporate governance standards across the sector. My analysis provides some empirical evidence to support the broad agenda of reforms put forth by the Board.

The rest of the paper is organised as follows. Section 2 provides context in terms of existing literature, while section 3 discusses the institutional framework in India. Section 4 describes in detail what a median board and director look like. The event studies are examined in 5. In section 6, I discuss construction of my bank risk variable, expand on the variables of interest, and present the results. Section 7 provides ideas for further exploration and concludes.

2 Related literature

The focus on top management, particularly the CEO, is motivated by the upper echelons theory (Hambrick and Mason, 1984). It states that the experiences, beliefs, and backgrounds of top executives affect their choices, and through their choices, the performance and strategic direction of an organisation (Wang *et al.*, 2016). Building on this, there exists a large body of literature on corporate governance, but a smaller and more recent one on corporate governance in banks after the global financial crisis, and an even smaller subset specifically on emerging economies.[7] Existing studies relate bank performance to board size (Adams and Mehran, 2011; de Andres and Vallelado, 2008), share of independent directors (Adams and Ferreira, 2007), CEO characteristics (Falato *et al.*, 2015; King *et al.*, 2016), CEO turnover, diversity (Adams and Ferreira, 2009; Ferrari *et al.*, 2017), and compensation (Chen *et al.*, 2006; Fahlenbrach and Stulz, 2011; Anginer *et al.*, 2016; Ongena *et al.*, 2018, among others).

[6]The BBB has no role to play in selection of non-official directors (which may be implicitly political in nature and difficult to track) or lateral hires. However, even though the process is more transparent and communicated via the Board's website, its recommendations are not binding and the Ministry of Finance still has the final say in who is selected. There is substantial evidence in the press since 2015 showing that often the Board's recommendations are not taken into consideration while making choices. See, for example, Bandyopadhyay (2017).

[7]I discuss the relevant literature in greater detail in each section of the paper.

My paper fits in the strand of research that links bank risk taking to corporate governance. It has been shown that boards with better corporate governance standards outperform their peers during crises. Aebi *et al.* (2012) show that US banks with more risk-management related corporate governance mechanisms in 2006 (such as the appointment of a chief risk officer) had higher stock market returns and return on equity during the financial crisis of 2007-08. Erkens *et al.* (2012) use information on financial institutions from 30 countries and show that firms with higher share of institutional ownership and independent directors took on greater risk prior to 2007. As a result, they suffered lower stock market returns during the crisis. Looking at the relative underperformance of state owned banks (measured as self-reported losses) in Germany between early-2007 and end-2009, Hau and Thum (2009) find that performance is correlated with financial expertise of the board members.

There are two papers closely related to mine. The first is Berger *et al.* (2014), which studies how the board's age, gender, and education effects risk-taking, defined as return to risk-weighted assets. Their identification rests on comparing banks facing retirement induced board changes in any given year to a similar bank that does not face any such changes.[8] They find that older and highly educated boards – measured as share of executives with PhDs – are associated with lesser risk-taking. Female directors in their sample of German banks self-select into stable and well-capitalised banks, but their appointment is linked to greater risk-taking in the following three years.[9]

The other paper closely related to mine is Sarkar and Sarkar (2018), which is the first of its kind study linking corporate governance and bank performance in India. The authors measure the effect of CEO duality and tenure, board size and independence, and share of nominee directors on bank performance (specifically, return on assets, non-performing assets, and market-to-book value). Their sample also covers listed banks, but over the period 2003–2012. They find that board size is insignificantly related to bank performance, while CEO duality, which was common in public banks up until 2015, has a negative effect on performance. They also find that board independence has a positive effect on performance of private banks, but the opposite effect in public banks. Longer CEO tenures are associated with improved bank performance. My work complements theirs along three dimensions. First, I provide a more detailed account of how board structures look across the two sets of banks, specially along less studied dimensions such as compensation. Second, in my empirical exercise, I am interested in bank risk-taking and linking that to the incentive structure of directors as measured by their gender, age, education, and compensation. Finally, the event study provides additional evidence on how shareholders respond to board turnover.

[8] This strategy requires a large number of banks in the sample, which is not possible in India, where only 38 banks are listed.

[9] There are a few studies directly linking governance with lending decisions. For example, Faleye and Krishnan (2017) construct a composite board effectiveness measure using board size, share of independent directors, CEO-chairperson duality, and board classification. They use this and information on bank-firm lending information to show that a change from first to third quartile of the board effectiveness measure is associated with a 7.1% increase in the probability of originating less risky loans.

3 Institutional framework and corporate governance rules in India

The governance framework is different for public and private sector banks in India. The Reserve Bank of India (RBI) is the regulator for domestic and foreign private banks.[10] On the other hand, corporate governance in public sector banks is governed both by the Ministry of Finance – under the respective legislations under which they were set up – as well as the RBI, which continues to have some room to make recommendations in certain specific aspects of the boards.[11] The Nayak Committee Report (2014) identifies the dual regulation of public sector banks as a major external constraint that currently prevents a level playing field between private and public sector banks.

In addition to these acts (i.e. wherever they do not conflict with the respective statutes of these banks), there are certain corporate governance norms which are enforced for all publicly listed firms by the securities markets regulator, Securities and Exchange Board of India (SEBI) (SEBI, 2000). Examples of these provisions include those on the presence of independent directors, minimum number of board meetings, board processes, and disclosure requirements, among others.

Indian bank boards operate under the single-tier system, like in US, UK and Canada, wherein the board handles both the management and supervision tasks and therefore consists of a mix of executive and non-executive directors.[12] This is in contrast with countries like Germany which have a two-tier system, where a board of directors that manages the day-to-day operations of the company is complemented by, and separated from, a supervisory board that monitors these decisions on behalf of other parties, like shareholders (Block and Gerstner, 2016; Adams and Ferreira, 2007).

A bank board is typically headed by the chief-executive office and managing director (CEO/CMD), who is the top decision maker of the bank. She handles strategic decisions with the aim of maximising shareholder value, and is accountable to the board of directors for the bank's performance. All executives report to the CEO and the CEO reports to the board. The chairperson is the head of the board of directors (which is elected by shareholders and responsible for protecting the investors' interests). She monitors the activities of the CEO but plays no role in day to day management. In most private banks, the positions of CEO/CMD and chairperson are carried out by two separate individuals; however, in most PSBs, the CEO/CMD was required by law to undertake both responsibilities until 2015 (*Indradhanush plan 2015*).

In the remainder of this section, I outline the relevant regulations for private and public sector banks (PSBs) with regards to two specific aspects that I will use in my analysis: hiring, firing, and retiring, and remuneration standards. The information below is drawn

[10] Private banks are governed under the *Companies Act, 2013* (henceforth, CA) and the *Banking Regulation Act, 1949 (henceforth, BR).*

[11] The relevant acts for public sector banks are: *Banking Companies (Acquisition and Transfer of Undertakings) Act, 1970* and *1980*, as well as the *State Bank of India Act, 1955*.

[12] An executive director is someone who has executive powers in the overall management of the company. A managing director or CEO, for example, is an executive director. An independent director on the other hand is a non-executive director. The chairperson of the board can be either. A non-executive director is not involved in day-to-day running of business but monitors the business activity. There are two types of non-executive directors - independent and non-independent.

primarily from the relevant acts, circulars, notifications, and press releases, as well as IMF (2018).

3.1 Hiring, firing, and retiring

According to the Nayak Committee Report (2014), public bank boards cannot be considered "independent" given the role played by the government in hiring most of the board (except shareholder-elected directors). Executive vacancies on public sector bank boards are typically filled by senior government officials, with the appointment carried out by the Ministry of Finance. Particularly for the chief executive officer (CEO), there are several restrictions on age and tenure, and the most important criteria for appointment is number of years spent in government service, or seniority (Sarkar *et al.*, 2019).[13] All appointments need to be cleared by the Central Vigilance Commission (CVC), which often creates delays if the appointment process is not started well in advance of the incumbent retiring.[14] Importantly, the outgoing CEO has no role in the appointment of her successor, and new appointments are always announced very close to the exit date of the incumbent CEO (Sarkar *et al.*, 2019). The tenure of the incoming CEO is usually fixed at three years, or retirement (also called "superannuation"), whichever comes first.[15] For private sector banks, there is no restriction on CEO age or tenures, except that it needs to be cleared by the central bank on a case-by-case basis.[16]

Since 1998, the retirement age in India has been fixed at 60 years, which all public sector employees are subject to. Most CEO/CMDs of state banks end up being hired close to their retirement, giving them tenures that are on average close to three years (Nayak Committee Report, 2014). There is no evidence that CEO/CMDs of public sector banks are removed on the basis of their bank's performance. Sarkar *et al.* (2019) show that CEO turnover in public sector banks is not linked to standard measures of bank performance, such as non-performing assets, loan loss provisions, loans, or net profits.

All board members need to pass certain "fitness and propriety" standards in order to be considered eligible. For example, CEO/CMD needs to be from the fields of business, finance, economics, or banking, and 51% of the board members need to have experience in specific fields like accounting, law, economics, agriculture, or small and medium enterprises.[17]

[13] For an example of a Ministry of Finance call for applications for board of directors of five PSBs in 2015, please see here.

[14] The CVC is a federal agency which is in charge of investigating alleged corruption and fraud in the government.

[15] The tenure term used to be five years or retirement, or whichever is earlier, up until 2014, and has been made *de-facto* 3 years or retirement (or whichever is earlier) since 2015. The appointments always contain a clause that allows the government to terminate appointments before the tenure is completed.

[16] There is evidence of the RBI stepping in to stop automatic renewals of CEO terms for private banks; see, for example, the case of private sector bank IndusInd Bank in 2014. In 2018, when the tenure for private lender YES bank's CEO was limited, the stock valuation of the bank fell by 34%, as can be seen here.

[17] The CEO also needs to declare any outstanding loans from the bank they manage, any defaults on these loans, as well as interests in other companies. There are some restrictions on directors holding positions on boards of other financial and non-financial companies. According to the listing norms, the director cannot be a member on more than 10 committees, or act as a chairperson of more than 5 committees across all companies in which he is a director (SEBI, 2000).

3.2 Remuneration standards

Compensation standards for public banks are based on government personnel rules which are consistent across all state banks (Sarkar and Sarkar, 2018). Within a board (public or private), there are different remuneration structures for executive and non-executive board members.

The total remuneration of senior managers in public sector banks – CEO/CMD, chairperson, and other executive directors – is comprised of two parts. The cash component, or salary, forms a bulk of the total pay. Since 2007, there has been a performance-linked incentive scheme for whole time directors of state banks which operates via the remuneration committee of the board.[18] However, the scheme is based on qualitative information and guidelines issued by the Ministry of Finance, and therefore, applies uniformly across state banks.

Specifically, banks are scored based on a performance evaluation matrix (with a weight 70%) that includes a host of financial variables that the state bank as an agent of the government needs to target. These include, for example, target growth rates in deposits, total lending, agricultural loans, and loans to weaker sections and small and medium enterprises. It also includes target amount of gross NPAs, but these typically show up on the balance sheets with a lag. The qualitative grading of senior management carries a weight of 30%.

On the other hand, in private sector banks senior management is incentivised on the basis of bank profitability, and a part of compensation is paid out through stock options. In practice, the RBI also uses its discretion to approve differential rates of compensation to whole-time directors of private banks (Nayak Committee Report, 2014), based on criteria such as bank size nature of bank operations, number of branches, and CEO characteristics. In general, the average compensation for top management is much lower in public banks as compared to private banks (more details in section 4).

A particularly interesting component is the sitting fees, which is a lump-sum amount of money paid to a director for each meeting attended by her. This is fixed internally within each private bank, and fixed across the board by the government for public sector banks.[19] All non-executive directors (possibly other than the chairperson) receive remuneration only in the form of sitting fees for each meeting of the board and its various sub-committees. The CEO/CMD and other executive directors do not receive sitting fees, nor do government of India and RBI nominee directors. No stock options are granted to any of the non-executive directors.

[18] Across all state banks, the remuneration committee is comprised of the RBI and government of India nominee directors and two non-executive directors.

[19] The government may change the sitting fees norms, but these would be applied across all banks in the public sector. The sitting fees may be different for the central board meetings and the other board meetings, but again this standard is consistently applied.

4 Characteristics of Indian banks: Public vs. private

There are approximately 187 banks in India, out of which 21 are public, 21 are Indian private, and 43 are foreign private.[20] The rest are co-operatives, regional rural banks, small finance banks, or payments banks. I have a smaller sample, however, as I focus on *listed* public and private commercial banks between 2006 and 2017, because these banks are required to make corporate governance disclosures. The dataset consists of 35 banks per year (figure 3). The Indian banking industry remains heavily regulated and has not witnessed a number of mergers or acquisitions in recent years (table 1).[21]

I have information on a cumulative 1356 directors over the time period 2006-2017 in *NSEInfoBase*, translating to an average of 357 active directors per year.[22]

Overall performance

In India, public sector banks are on average larger than private sector banks: their loans outstanding, risk-weighted assets, total income, and non-interest income are nearly twice as large as their private sector counterparts. However, state banks fulfil an important social welfare objective, as their lending to the priority sector is nearly thrice as large as compared to the private sector.[23]

Between 2006–2017, public banks have less healthy balance sheets. On average, private banks had 2.86% gross NPAs as share of total advances, whereas public banks had nearly twice that (5%). State banks have on average 0.6 to 0.8% less net interest margins than private banks; are less capitalised; have a return on equity of 8.43% as compared to 13% at private banks; earn lesser profits than private banks; and on average have a price-to-book ratio of 0.95 as compared to 2.21 at private banks.

I discuss heterogeneities within the two groups of banks in the remainder of the section. Table 2 contains information on specific corporate governance characteristics of the median director (in panel A) and median bank (in panel B) in my sample.

Board size, age, and diversity

The median bank's board of directors comprises or approximately ten members.[24] The median age of an active director is 58 years and an inactive director (that has left the bank) is 59 years. The age statistics are important because the retirement age for public sector banks is set at 60 years. The average length of tenure is approximately 3.2 years, but this is driven primarily by public banks.

[20] For more, see https://www.rbi.org.in/commonman/English/Scripts/BanksInIndia.aspx.

[21] There are only three mergers between 2006 & 2017, which is accounted for in my data.

[22] As a robustness check, I also use *CMIE Prowess* to double-check whether information on board members matches and find that to be the case.

[23] In India, priority sector lending includes lending specifically to: agriculture, micro, small and medium enterprises, export credit, education, housing, among others. Every year, targets are set for all banks in the country, but those for public banks are more binding. See, for example, here.

[24] According to the Companies Act (2013), each listed company must have between three to fifteen directors, and allows for a special vote to appoint more.

Private sector banks have older boards on average as compared to public sector banks. For example, a median director leaving a private sector board is on average 3.2 to 4.3 years older than those on a PSB board. This is more a feature of the strict retirement policy that state banks are subject to (but not private banks).[25] Correspondingly, directors on state bank boards have shorter tenures of just 2.25 to 3.2 years.[26] The median CEO of a state bank has a tenure of just 2 years (echoing the findings of Nayak Committee Report, 2014), whilst the median private sector CEO has a tenure of five years.

It is worth mentioning that unsurprisingly the median board of directors has low diversity in terms of gender, with women making up only about 6% of directors on average. However, public sector banks do far better in gender diversity than private sector banks, as shown in table 2. The Companies Act (2013) mandated that all listed companies in India have at least one woman director on its board, and required that all vacancies for the seats of women directors be filled within 6 months. Consequently, the situation has been improving in the second half of the sample, with 28 out of 38 banks having more than the median (8%) of their boards comprised of women in 2017.

Board effort and compensation

The median bank has roughly 12 sub-committees to carry out the work of the board.[27] The median public sector bank board member is busier *within* the bank, as she is a member of a higher number of sub-committees. She also attends more board meetings as compared to her private sector counterparts. These include, but are not limited to, committees on non-performing assets, human resources, remuneration rules, risk, and audit. In particular, I am interested in the meetings of the risk committee, which oversees integrated risk management policies, relating to credit, market, and operational risks. As shown in table 2, there are more directors on the risk sub-committee for public banks than private ones, although this may simply be a function of the larger board size of public sector banks.

The median director on a public sector bank board earns only about one-third of her private sector counterparts ($15,000 per year vs. $50,000 respectively).[28] This disparity continues and worsens at higher levels of management as well – public sector CEOs make only a tenth of their private sector counterparts' total remuneration (in real terms).[29] Figure 4 shows the salary (i.e. cash) and variable components as a share of total remuneration for all CEOs and chairpersons in the sample. On average, salary is the dominant form of compensation for these two senior positions, and the share is even higher for public banks as compared to private ones.

[25] The oldest director in the sample was noted agricultural scientist Dr. MS Swaminathan, the father of India's "green revolution" and an ex-minister from the agriculture department. He served 2 years as an independent, non-executive director at the State Bank of India, the largest public sector bank in India.

[26] The person with the longest tenure of 25 years in the sample is Mr. Anand Gopal Mahindra, one of the members of the family that founded the private Kotak Mahindra Bank. He was an independent director.

[27] The maximum is 22 for Andhra Bank in 2015 and Punjab National Bank in 2016 (both of which are state banks).

[28] All compensation variables are converted to real 2015 Indian Rupees, and then converted to US dollars. For conversions, I use a USD-INR rate of Rs.69.

[29] The highest paid individual in the sample, earning $1.37million a year, is the MD of HDFC Bank, the largest bank in India by market capitalisation.

Monitoring

Private bank boards have a higher share of independent directors on their board over the entire sample, 57.14%, as compared to the median value of 0% for state banks. However, there has been a substantial increase in the share of independent directors at public sector banks after 2013.

Next, I calculate the extent of "soft" influence that the CEO has over the board by using the share of current board members that were appointed after the serving CEO in figure 5. Clearly, private sector bank CEOs (right panel) have higher power over the board by this measure because of their longer tenures. However, interestingly, CEO power over the board is also not very low for public sector bank boards (left panel), most likely because all directors are subject to the same retirement policies and there is high turnover in general. Actually, the data more than likely underestimates the soft power of the state bank CEO, given that she is a senior bureaucrat from the government, and the board members are other public sector employees.

There have been very few performance-based removals of board directors. In the sample, only 2% of all directors have ever been suspended, terminated, or superceded. Directors most often leave their position because they either retire or their term expires (table 3). There is not much heterogeneity between the two bank groups in terms of reasons why directors leave their positions, except that the end of nomination is a much more important reason for public bank boards than private ones.

5 Are changes in the board "news"?

My next research question is whether changes in the composition of a bank's board are considered as "news" by shareholders. I investigate this by conducting an event study of abnormal stock market returns around days when board changes are made, using information on the appointment and cessation dates.[30]. In particular, I am interested in chief executive officers (CEOs) and independent directors.[31]

The event, which is defined as the day when director turnover takes place, is designated as 0; therefore, the day before the event is (-1). I first estimate a market model as shown in equation 2 over an estimation window of 80 days (i.e. $-91 : -11$) before the event:

$$R_{bt} = \alpha_b + \beta_b R_{mt} + \epsilon_{bt} \tag{1}$$

for bank b on day t. R_{mt} is the market return. The market model helps to strip bank-wise returns of any market-wide effects, for example, due to a crisis, budget announcements, elections, or festivals. I can then calculate bank-wise abnormal returns (ARs) using $\widehat{\alpha}_b$ and $\widehat{\beta}_b$ estimated from equation 2 for each day in the event window as follows:

[30]I use the term cessation interchangeably with resignation, and interpret it broadly to refer to all reasons for leaving shown in table 3

[31]I manually cross-checked a random sample of CEO appointment and resignation dates, and found that most of them are announcement dates. However, in cases when the actual announcement dates cannot be verified, the implementation dates are used.

$$AR_{bt} = \text{actual return}_{bt} - \widehat{\alpha}_b - \widehat{\beta}_b R_{mt}, \forall\ t = (-1, 0, +1, +2) \tag{2}$$

In the next step, I rebase the abnormal return on day (-1) in the event window equal to 0 for ease of interpretation and comparison, and then cumulate the abnormal returns over the next 3 days. Finally, I average the cumulative abnormal returns (CARs) for each day across all banks, and use bootstrap inference to derive the 95% confidence intervals and plot them. Using a very short window around the event date (-1, +2) allows for cleaner identification of the effects as it reduces the likelihood that the results are driven by confounding factors (Gürkaynak and Wright, 2013).

5.1 CEO turnover

The question of whether board turnover, particular CEOs and independents, matters for stock market performance, has been studied before (Bonnier and Bruner, 1989; Huson *et al.*, 2004). A closely related paper to mine is Pessarossi and Weill (2013), who study the effect of CEO turnover on firm value in China.[32] The authors cite three potential channels due to which stock markets may respond to CEO turnover. First, firm value may increase in response to CEO turnover if the incoming CEO signals an improvement in management over the incumbent (ability hypothesis). On the other hand, stock markets will respond negatively if investors believe that turnover is likely to be associated with revelations about the true extent of underperformance under the incumbent (information asymmetry hypothesis). However, it may also be that if all managers are of similar quality and the controlling shareholder only uses dismissal as a credible threat in the event of underperformance, then turnover implies no improvement in board quality (Kim, 1996, for example). Therefore, there will be no response of stock markets to turnover (scapegoat hypothesis). They find positive and significant market reaction to CEO turnover announcements, which is consistent with the ability hypothesis. It implies that the market anticipates a future increase in firm performance after a CEO turnover.

There are 120 unique CEOs in my dataset, of which 36 are active at the end of the sample in 2017.[33] Table 4 shows that of the CEOs that exit the sample, roughly 70% do so because they either retire, are superannuated, or because their term at that bank expired and was not renewed.[34] As discussed earlier, performance based removal of CEOs is uncommon, especially in the public sector.

To construct the set of events to be used in the event study, I get rid of events where I do not have information on the CEO's leaving date, leaving reason, gender, or education. I also only use events after 31 March 2003 to allow for more listed banks. This gives 108 appointments and 76 cessations, for a total of 184 events. I first pool all the events (both appointments and cessations) and conduct the event study on these events.

[32] China would be a close match to India in terms of the role of the state in the public sector, which gives rise to the kind of corporate governance constraints that I study here.

[33] There should be 38, but there are vacancies in 2 banks as of 31 March, 2017.

[34] Note that there are 50 CEOs in the dataset who switch between banks in the sample (for example, as an executive director of one bank, and then CEO of another), of which at least 6 are moved to other banks as CEOs. This is why the total number of events in table 4 is more than 84.

For the baseline specification, I use the *Nifty 50* as the market index.[35] Figure 6 shows the baseline event study for all pooled CEO turnover events. The black solid line is the average effect, while the dashed lines around it are the 95% confidence intervals. For all banks in panel (a), CEO turnover is associated with a +1.3% cumulative abnormal return after 2 days of the announcement, which is significant at the 5% level. Splitting this up by type of banks, we can see that it is public sector banks driving the results: from panel (b), we can see that the public sector cumulative abnormal return (CAR) is even higher at +1.74% (also significant at the 5% level). On the contrary, for private sector banks, there is a slight negative effect at announcement (i.e. from day −1 to day 0), which doesn't change much through the event window and becomes insignificant by day +2.

All the results are robust to using an alternate market index, the *Nifty Bank*, as shown in figure A.1.[36] They are also robust to using an alternate estimation window of (−91, −2).[37]

If appointments and cessation are treated as different types of "news" by shareholders, then pooling them or averaging across them will give us a muted effect. Therefore, next, I split the events both by bank, as well as by the type of event, to see whether shareholders are sensitive to the type of CEO event. It is possible to contrast these events because resignations and appointments are not perfectly aligned, i.e. it is plausible for there to be a vacancy period even for the CEO position before a replacement is announced. In fact, one of the key recommendations of the Bank Boards Bureau for public sector banks has been to prevent such unintended vacancies. Additionally, there may be a lag between announcement and implementation of the personnel change.

In figure 7 panels A.1 and A.2, I report the same event study separately for CEO appointments and resignations respectively, and find that the cumulative abnormal return over the event window is in the same direction and of roughly similar magnitude across the two types of events. This is particularly true for public sector banks in panels B.1 and B.2, where CEO appointments give a CAR of +1.86%, and CEO resignations give a CAR of +1.6%. This tells us that any kind of turnover in the CEO is understood as positive news from the point of view of a public sector bank's shareholders. On the other hand, CEO appointments and resignations for private sector banks are considered as insignificant "negative" news by the bank's shareholders, with CARs of −1.5% and −0.42% respectively. Particularly for appointments, there seems to be a significant negative reaction on the day of the announcement to the tune of −2%, which dies out over the next two days.

Overall, I find evidence that is consistent with Pessarossi and Weill (2013) for China. In general, shareholders anticipate an improvement in future bank performance in response to CEO turnover, possibly due to an improvement in managerial quality. The effect is stronger for public sector banks than for private ones.

[35]The *Nifty 50* index is the benchmark broad based stock market index for the Indian equity market. It is based on 50 of the largest and most liquid Indian securities, capturing about 67% of the free-float capitalisation of the stocks listed on NSE as on March 29, 2019. More information is available at `https://www.nseindia.com/products/content/equities/indices/nifty_50.htm`.

[36]The *Nifty Bank* is a benchmark index for the banking sector. Its is comprised of the 12 most liquid and large capitalised Indian banking stocks. More information is available at `https://www.nseindia.com/products/content/equities/indices/sectoral_indices.htm`.

[37]Not shown here in the interest of space.

5.1.1 Explaining CEO turnover CARs for public banks

It is interesting also to determine what drives CARs for public sector banks, especially with respect to CEO and bank characteristics. I focus on public banks, since abnormal cumulative returns at the end of the event window for private sector banks are insignificant. Therefore, I run the following cross-sectional ordinary least squares (OLS) regression, where d is director, b is bank, and t is the event date:

$$\begin{aligned}\text{CARs}_{dbt}^{(-1,+2)} = {} & \alpha + \beta_1 \text{CEO age at appt}_{dbt} + \beta_2 \text{CEO gender}_{dbt} + \beta_3 \text{CEO education}_{dbt} \\ & + \beta_4 log \text{ assets}_{dbt-1} + \beta_5 \text{return on assets}_{dbt-1} + \beta_6 \text{capital surplus less } 8\%_{dbt-1} \\ & + \beta_7 \text{net NPA}_{dbt-1} + \beta_8 \text{board size}_{dbt-1} + \beta_9 \text{number of meetings}_{dbt-1} \\ & + \epsilon_{dbt} \qquad (3)\end{aligned}$$

and where $CARs$ are the cumulative abnormal returns over a three-day window around the event date. The setup of my specification is similar to MacKinlay (1997); Nguyen and Nielsen (2010); Andries *et al.* (2019). All the bank specific variables are at the end of previous March and standard errors are clustered by director.[38] There are three CEO characteristics that are of particular interest: CEO's *age* at appointment, *gender*, and *education*. Age at appointment has information both about the likely tenure, as well as past experience, of the CEO. Shareholders may penalise older CEOs since they will stay on the job for a shorter period of time, but may reward the CEO's past work experience.

The CEO's gender might also play an important role, but its relationship with stock market returns is ambiguous. Female CEOs may be able to leverage unique skills and different networks, and if they are perceived to be less aggressive risk-takers than their male counterparts, then they may create shareholder value since future profitability will be higher (Nguyen *et al.*, 2015). Recent work by Faccio *et al.* (2016) has shown that firms run by female CEOs, as compared to those run by male CEOs, tend to have less volatile earnings, lower leverage, and higher chances of survival. On the contrary, if female CEOs undertake excessive monitoring, they may decrease shareholder value (Adams and Ferreira, 2009). If female CEOs have lower job experience, their appointment may also be associated with negative firm value as Ahern and Dittmar (2012) find for Norway. In particular, Martin *et al.* (2009) find that the three-day cumulative abnormal returns in response to female CEO appointments are not significantly different from those with male CEO appointments. Similarly, Wolfers (2006) does not find any systematic differences in stock returns in female-headed firms. On the other hand, Lee and James (2007) observe negative abnormal returns for women CEO appointments, but find that women who are promoted from within a firm are viewed more positively. There is limited research in banking specifically, but Nguyen *et al.* (2015) also do not find that gender of CEOs is linked to abnormal stock market returns.

Higher education attainment may increase an individual's propensity to take risk in financial decisions. Falato *et al.* (2015) argue that boards use CEO credentials as publicly-observable signals of otherwise unobservable or hard to quantify CEO skills, and find

[38]Therefore, an appointment at 31 July 2006 is merged with bank performance at 31 March 2006; similarly a resignation at 07 March 2006 is merged with bank variables at 31 March 2005.

that CEO credentials are linked to better firm performance.[39] In a meta-analysis of 308 studies on the topic, Wang *et al.* (2016) find that a CEO's formal education is important for the firm's future performance, because it is directly linked to how she handles complex information and makes decisions. Nguyen *et al.* (2015) use Ivy-league education and specialisation (proxied by an MBA degree) as their measures of CEO education and find that it is positively linked to shareholder value creation.

I proxy education using four variables. $Masters$ is a dummy that takes value 1 if the CEO has a masters degree; similarly for PhD. *Relevant* is a dummy for six "relevant" degrees or qualifications, namely, chartered accountancy, commerce, economics, management, MBA, or membership of the Indian Institute of Banking and Finance.[40] MBA is a dummy for an MBA or other management-related qualifications or certifications.

Table 6 shows the results of this exercise. Out of the CEO characteristics, interestingly *gender* is positive and significant throughout at either 5% and 10% levels, depending on the specification. Therefore, female CEOs in the public sector are associated with higher shareholder value. [41] However, this result may be driven by the fact that entry requirements for female CEOs are simply higher, implying that the women who do get hired are more qualified than the average male CEO. *Age* is negatively associated with abnormal returns in the public sector, consistent with the idea that older CEOs that are likely to hit the retirement age ceiling sooner, are viewed negatively from the shareholder's perspective. However, the variable is insignificant throughout. CEO *education* is always positive but not significant – therefore, abnormal returns are technically higher for more educated CEOs.

In bank characteristics, there is evidence that abnormal returns for CEO turnover are particularly lower for more profitable banks. This implies that if a bank has a higher return on assets, a change in the CEO is viewed negatively from the perspective of shareholders, since it is unclear how the new individual at the helm will drive future profitability.

Another interesting implication is that turnover at banks with higher net non-performing assets is associated with *lower* abnormal returns. There are three hypotheses from the earnings management literature which potentially help explain this finding. The "big bath" hypothesis (Pourciau, 1993) argues that incoming CEOs resort to window-dressing accounts in their transition quarters in an effort to show higher profits at the end of their tenure (due to base effects). According to the "truth-telling" hypothesis, new CEO's may be more inclined to reveal the true extent of underperformance under the outgoing CEO in an effort to pose themselves as effective monitors (Hertzberg *et al.*, 2010). Finally, as per the "personal risk management hypothesis" (Amihud and Lev, 1981), the incoming CEO may increase provisions and reduce lending in an attempt to minimize any personal

[39]"Credentials" are measured as publicly observable signals of CEO education and professional records, as well as reputation, proxied by coverage in the business press (Falato *et al.*, 2015).

[40]The *Indian Institute of Banking & Finance* is a "professional body of banks, financial institutions and their employees" in India. To be a member, an individual only needs to be an employee of a financial institution. However, members do have the opportunity to access several banking and finance related-courses through the Institute, as well as professional training programs. See, for example, `http://www.iibf.org.in/aboutus_history.asp`.

[41]Ferrari *et al.* (2017) use the exogenous implementation of gender quotas for board of listed companies in Italy. Using event studies, they find that announcement of the reform was linked to positive abnormal stock market returns.

costs from negative outcomes due to the incumbent's actions.

Sarkar *et al.* (2019) find evidence consistent with the personal risk management hypothesis for CEO turnover in Indian public sector banks. They also show that incoming CEOs increase provisions for bad loans by 8.5% in the first quarter in which they take charge. Therefore, this in the context of my results would indicate that the higher the NNPAs of the bank, the more likely the incoming CEO will provision and cut lending at a higher rate, thereby impacting profitability. Note that all three hypotheses mentioned above work in the same direction for my results.

5.2 Independent director turnover

I now turn to a study of whether shareholders respond to announcements regarding independent directors. Independent directors, who have no direct ties with management, may be considered more effective monitors of management because they are technically not answerable to management; however, they may lack firm-specific knowledge, which may turn out to be quite costly if regulation limits the pool of eligible directors. Therefore theoretically the relationship between the share of independent directors and bank value may run either way. There are a few papers related to what I do. In particular, Rosenstein and Wyatt (1990) find that the appointment of an outsider as an "independent" director is accompanied by significant positive abnormal returns. The authors argue that these results are consistent with the idea that outside independent directors are selected to maximize shareholder wealth. Nguyen and Nielsen (2010) study sudden deaths of independent directors for US firms between 1994 and 2007. They find that director death is associated with a stock price decline of 0.85% on average.

There are 658 independent director announcements in my sample, of which 71% are appointments. Out of the 193 events where independent directors leave the board, 109 are because of retirements, superannuation, or term expiry (similar to what I found for CEOs). I run a similar event study as before, using the *Nifty 50* as the market index and $(-91, -11)$ as the estimation window for my baseline specification. However, I have much less information on individual characteristics of the independent directors, such as their education. Nevertheless, I omit directors with no information on their date of birth and use events after 31 March 2003 as before. This exercise yields a smaller set of events: 356 appointments and 168 resignations. The median tenure of an independent director on a public bank board is unsurprisingly much shorter (3 years) than that on a private bank board (7 years).

The results are presented in figure 8. I find that independent director appointments for Indian public sector banks are shareholder value creating: the cumulative abnormal stock market return increases by +1% after 2 days (significant at 5% level). On the other hand, cessations are associated with negative returns.[42] For the private sector, announcements on independent directors are not significantly different than zero. This seems to indicate that shareholders of public banks value independent directors – and their ability to effectively monitor the board – positively.

[42]Here, using the alternate market index, *Nifty Bank*, shows a sharp fall in abnormal returns the day after the announcement, i.e. at +1.

6 Bank risk and board characteristics

We now turn to the linkages between bank risk and board characteristics. In this section, I first describe my main dependent variable, the z-score, then discuss the model and explanatory variables of interest, and end with a description of the results.

6.1 Constructing the z-score

I measure bank risk using the z-score of each bank b, similar to Laeven and Levine (2009), shown below:

$$z - \text{score}_{bt} = \frac{\text{ROA}_{bt} + (\text{K}/\text{TA})_{bt}}{\sigma_{bt}^{ROA}} \tag{4}$$

where ROA is return on assets, K is total equity, TA is total asets, and σ^{ROA} is the sample standard deviation of the return on assets for bank b at time t. The z-score measures the distance from insolvency for the bank and indicates the number of standard deviations that a bank's return on assets has to drop below its expected value before equity is depleted. Therefore, a higher z-score means a bank is more stable. I have 33 observations where the z-score as defined by equation 4 is negative. Most of these observations are concentrated in public sector banks in 2016 and 2017 (there are also two private sector banks). All banks with negative z-scores have positive capital but negative average return on assets.

Negative z-scores reflect a drawback of this measure as applied to public sector banks: even if the z-score for state banks is low or negative, the actual risk of default may still not be high because the banks are implicitly guaranteed by the state. I still prefer this metric because it gives an intuitive, relative ranking of bank stability, allowing me to compare public and private sector banks on the same pedestal. However, for the purposes of the technical interpretation, I translate the z-score for state banks as likelihood not of default, but of "recapitalization".

In my sample, the z-score ranges from -1.84 to 21.92, with a median equal to 2.2. Figure 10 shows the density plot of the z-score in 2006 and 2017 for the entire sector. The left shift of the distribution in 2017 demonstrates that the banking sector has become riskier as a whole. The average public bank (1.67) has one-third the z-score of a private bank (4.98) in the sample. This means that profits would have to fall by a little over 1.67 standard deviations to completely deplete the average state bank's equity, though they would likely receive government help long before that point is reached. Of course, part of the significant difference between the default risk of these two types of banks has to do with the kind of business models and lending they do. For example, public banks are the dominant source of credit to corporates, and also have a higher share of lending for social welfare reasons, such as to priority sectors like agriculture, whereas private banks are more retail lending oriented (IMF, 2018). Wherever data is available, I will control for these differences in business models.

6.2 Empirical specification

The baseline specification for bank b at time t is as follows:

$$\begin{aligned} log\ z - \text{score}_{b,t} = \alpha_b + \gamma_t + \boldsymbol{\beta_A}\text{board variables}_{b,t-1} + \boldsymbol{\beta_B}\text{board variables}_{b,t-1} \times \text{D.public}_b \\ + \beta_1 log \text{ total assets}_{b,t-1} + \beta_2\text{net NPA/loans}_{b,t-1} + \beta_3\text{NII}_{b,t-1} \\ + \beta_4\text{NIM}_{b,t-1} + \beta_5\text{loans/assets}_{b,t-1} + \beta_6\text{sensitive sector credit/assets}_{b,t-1} \\ + \beta_7\text{non-resident FII}_{b,t-1} + \epsilon_{b,t} \end{aligned} \quad (5)$$

where NII is non interest income, NIM is the net interest margin, and FII is foreign institutional investors. The coefficients of interest, β_A and β_B, are identified by within-bank variation. Standard errors are clustered at bank-level. The main board variables of interest are CEO and board age, tenure, compensation (and its components), as these are more exogenously determined for the public sector. I am also interested in meetings of the risk committee, average board size, share of women on the board, and share with relevant education.

Tenure and age

The first board variable of interest is the CEO and board's *tenure* and *age*. Tenure may be an important influencer of bank risk. Longer-tenured CEOs, in particular, tend to initiate fewer strategic actions (Wang *et al.*, 2016) and their tenure may be considered a proxy for risk aversion (Ongena *et al.*, 2018). Further, CEOs with longer tenures are more likely to be entrenched, and have significantly higher power over the board (Coles *et al.*, 2006), since most of the directors are hired after them. These boards may therefore end up being less efficient in their monitoring tasks (Aebi *et al.*, 2012; Nguyen *et al.*, 2016).[43] On the other hand, very short tenures may reduce the efficiency of senior management, allowing them less time to learn on the job.

Applying this to the Indian case and using the descriptives developed in section 4, I hypothesise that CEO tenures for public sector banks are likely too short. Coupled with the fact that the median CEO retires after completing her tenure, she has lesser incentive to monitor the bank diligently as compared to a CEO who has to stay on for longer. This means that an increase in the tenure should be associated with better monitoring and more time to develop strategy. Therefore, for these banks CEO tenure should be associated with lower bank risk.

I also need to control for the director's age. As discussed in Berger *et al.* (2014), risk-taking behaviour usually declines with age. For example, Hambrick and Mason (1984) find that firms headed by younger CEOs tend to take more risk. This may be simply because more mature executives are more experienced, or they may be less concerned about job-security. In public sector banks in India, this channel may be of particular importance since most CEOs are career bureaucrats, and are usually hired close to retirement. I anticipate that on average CEO age is negatively linked to bank risk.

[43] In extreme cases, long tenures can lead to significant capture of the board, and allow suspicious activity to go undetected; see, for example, Nguyen *et al.* (2016).

As discussed earlier, in Indian public sector banks, the outgoing CEOs know their term limits while incoming CEOs get to know about their appointments just a month or so before assuming office (Sarkar *et al.*, 2019).

Total remuneration

The next board variable of interest are components of the CEO's *total remuneration*. While stock and options-based compensation can tether shareholder value and CEO wealth (Murphy, 1999), it can also aggravate the risk-shifting problem, where shareholders have an incentive to shift risk to bondholders. In less efficient markets, high sensitivity of wealth to stock or stock options can also incentivise CEOs to take a short-run view with respect to profit maximization by aggressively expanding business (Fahlenbrach and Stulz, 2011) and taking on additional risk. Chen *et al.* (2006) find a positive effect of option-based compensation on market measures of risk for US commercial banks between 1992 and 2000. On the other hand, Ongena *et al.* (2018) study risk-taking incentives of bank holding companies in the US and find that while the former is insignificantly related to bank risk in normal times, the relationship becomes stronger in a downturn. The authors also find some evidence of a moderating effect of pay-for-performance incentives on bank risk.

However, an important implication of this literature is that shareholder-friendly boards are associated with higher bank risk. For instance, Fahlenbrach and Stulz (2011) investigate whether bank performance during the crisis was linked to CEO incentives and find some evidence that US banks with CEOs whose incentives were better aligned with the interests of shareholders in 2006 had a worse share price performance during the subsequent crisis. Anginer *et al.* (2016) find that shareholder-friendly boards with conventionally "good" corporate governance strategies – such as board independence – can also be linked to lower bank capitalization and by extension higher bank risk. This channel is likely to be quite important for India.

In most Indian banks, a flat salary is paid only to the whole-time executive directors, which includes the CEO and other managing directors; occasionally the chairperson may also be paid a flat remuneration. CEO and other managing directors of private banks have additional components of their total pay – they may be paid allowances for housing, stock ownership plans, stock options, and performance bonuses – this makes up their "variable pay". For public sector banks, the salary contains all the allowances and the "variable component" is decided by the government of India based on a performance evaluation matrix, as discussed in section 3.2. Sitting fees are paid to all the non-executive directors, and potentially also the chairperson.

My hypothesis is that a higher share of salary (or cash component) for public sector banks should be associated with lesser risk. This is because the variable component which prioritizes expansion in business, in combination with shorter tenures, may directly incentivise CEOs to take on additional risk to maximize their wealth. However, they may also be motivated by reputation concerns. For private sector banks, I expect the reverse. If the CEO is also founder and owner of the bank, then she might be more risk-averse as compared to other CEOs who are agents of the owners. If she is not the owner, and is not bound by limits on tenure, then she may be able to take a long-run perspective to ensure that the bank is taking on optimal risk.

Other board variables

I will also be controlling for *board size*, *meetings of the risk committee*, *share of female board members*, and *share of directors with relevant education*, which are defined in section 5. There is ambiguity in existing literature on how these variables are linked to bank risk.

Board size is defined as the total number of active directors for that year and that bank. For Berger *et al.* (2014), who study bank risk and corporate governance, the number of other board members may act as a counterbalance to a powerful or entrenched CEO.[44] *Meetings of the risk sub-committee* are the total number of meetings for that sub-committee for that year. Aebi *et al.* (2012) use a similar variable as a control in their study of risk-related corporate governance mechanisms in bank boards in 2006 and the bank's stock market returns and return on equity during the financial crisis of 2007-08. I anticipate that the number of meetings should therefore be positively linked to bank safety if the bank is proactively and continuously managing its risks. On the other hand, if the bank only constitutes the committee or calls for more meetings once risks have materialized, then it may be negatively associated with bank safety. If the committees are not very effective, we may see no effects.

Share of female board members is defined as the total number of active female directors on the board. As before, the *share with relevant education* is the number of active directors who have qualifications relating to chartered accountancy, commerce, economics, management, MBA, or membership at the Indian Institute of Banking and Finance (IIBF).

Bank controls

Finally, I have a standard set of bank controls, which are widely used in the literature studying bank risk. The first is *log total assets*, because larger banks which are better diversified may be less risky. On the other hand, if public banks are larger, then they may be riskier from the z-score point of view since the business model revolves around fulfilling certain social objectives (for example, lending to agriculture or industries with long-gestation periods).[45] For proxying balance sheet composition, I use *loans to assets* and *credit to sensitive sectors*, where "sensitive sectors" are real estate, capital markets, and commodities. I have less complete data on *credit to agriculture*, but I use that in one of the robustness checks. I also control for the extent of diversification of the bank's balance sheet by using *non-interest income/assets* (NII).[46] To capture asset quality, I use *net-non performing assets/loans*, which is defined as gross NPAs less provisions, and to proxy for profitability, I use *net interest margins* (NIM). For ownership, I also include an additional variable on the investment by *non-residential foreign institutional investors*.

[44] There may occasionally be a trade off between large and small boards. Smaller boards can have more cohesion, lower decision-making costs (such as coordination or communication), and less likelihood of free-riding (de Haan and Vlahu, 2016; Jensen, 1993). Larger boards on the other hand may bring in more information and diversification of experience, as well as better linkages with subsidiaries, or other firms, therefore increasing performance; but may also lead to group-think, reduce information sharing, and give excessive control to the CEO, therefore reducing efficiency. Generally performance varies inversely with board size outside a narrow range, which is typically 10-12 members (Walker, 2009).

[45] Lending to sectors that are under-serviced by private banks might mechanically put downward pressure on the state banks' return on assets.

[46] To construct the series on non-interest income, I add the following components: interest from investment, export income, other income excluding ordinary income and extraordinary income.

6.3 Results

Before interpreting the results, it is useful to recap that because of the way the z-score is constructed, an increase in it means that a bank has become safer (because it now needs its ROA to fall by more to wipe out its capital). A decrease in z-score means that the bank has become riskier.

Table 8 shows the results regarding the board and CEO tenure and age. Columns (1) to (4) contain (natural logs of) board and CEO tenure and in columns (5) and (6), I further control for the average ages.[47] For columns (1), (3), and (5) instead of a full set of time fixed effects, I include fixed effects for two time periods: 2006 to 2011, and 2012 to 2017, which are based on the Indian business cycle. It also helps alleviate the issue of time persistence in board characteristics. Columns (2), (4) and (6) contain full time fixed effects.

From column (1) and (2), I observe that CEO tenure does not have a significant relationship with bank risk. From the second row, we can see that CEO tenure has a positive sign for public sector banks in all columns barring one, but it is significant at conventional confidence levels only in columns with reduced time fixed effects. In sum, there is evidence that higher CEO tenure for state banks is associated with an increase in bank safety. This is different from Sarkar and Sarkar (2018), who find that CEO tenure is associated positively with return on assets and that this result is primarily driven by private sector banks. However, the key difference between our specifications is that I include bank fixed effects instead of treating all public banks as a homogenous group. For private banks, CEO tenure is consistently negative but insignificant in all specifications barring one.

Board tenure seems to play a more important role. An increase in average board tenure in private banks and a decrease in average board tenure in public banks is associated with lower bank risk. Both coefficients are significant, at 1% and 5% levels respectively. This correlation does not go away even when I control for CEO age and board age in columns (5) and (6). On the other hand, CEO age in columns (3) and (4) has a positive sign: this implies that older CEOs, who by definition have more experience, are linked to lower bank risk. Columns (1) and (2) of table 9 contain the same specification but with additional information on the share of female directors and share with relevant education. The results on tenure and age do not change.

Next, I shift focus to relative values in column (3) and (4) of table 9, by defining two new variables, the relative age of the CEO with respect to the board, and the relative tenure of the CEO compared to the board. Here, I find evidence that an increase in the CEO's tenure with respect to the board's is positively linked to the z-score for public banks (making them safer), and negatively linked for private sector banks.[48] Both the coefficients are significant.

The controls have the expected signs. Specifically, a decrease in net NPAs and an increase in non-interest income and net interest margins is associated with lower bank risk. There is some evidence that investment by non-resident FIIs is linked to lower bank risk,

[47] I use logs for board and CEO tenure as these variables are right-skewed, whilst age is not. However, using *log age* does not change the results.

[48] The result that an increase in the CEO's tenure with respect to the board's is positively linked to the z-score for public banks is consistent with the earlier result that an absolute increase in the *board's* tenure is associated with a riskier bank.

potentially due to better risk-management policies.[49] None of the other board variables are significant.

In table 10, I test the hypothesis whether there is any link between CEO compensation and bank risk. For now, I use CEO salary to total remuneration in columns (1) and (2), and CEO total variable compensation to total remuneration in columns (3) and (4), where the latter is defined as follows (all compensation variables are in real 2015 Rupees):

$$\frac{\text{Total variable compensation}}{\text{Total remuneration}} = 100 \times (1 - \frac{\text{Salary}}{\text{Total remuneration}})$$

The table reflects that an increase in the salary component of a CEO's total remuneration is significantly (at 10%) negatively associated with bank risk at public sector banks, whilst the opposite is true for private sector banks. The correlation is significant and opposite in sign for the total variable component. In addition, I look not only at the last period's compensation, but also that of the previous CEO (since the median CEO tenure in the sample is 3 years). I find that the pattern holds, with both higher magnitudes as well as higher significance. This is consistent with my hypothesis that variable pay at public banks when combined with shorter tenures provides incentives to state bank CEOs to undertake higher risk. For private banks, variable pay aligns the CEOs incentives with those of the shareholders but the lack of constraint on tenures allows them to take a long-term view of profitability (Fahlenbrach and Stulz, 2011). Now the variable share with relevant education becomes negative and significant, indicating that boards with more qualifications in commerce, economics and related fields are linked to higher risk.

Next, I investigate the absolute values of total variable compensation, salary, and total remuneration. One issue here, which wasn't a concern with shares used above, is that larger banks may pay more simply by virtue of being larger. One reason for this could be that larger banks require more effort from their directors, and hence, compensate accordingly. This is certainly likely to be true for private sector banks, although not necessarily for public sector banks.

Therefore, I first regress CEOs total remuneration, salary, and total variable pay (in real 2015 Indian Rupees) on total assets, and three CEO characteristics: relevant education (a dummy), meetings attended, and age for each bank b and time t:

$$\begin{aligned} log\text{ CEO pay}_{bt} = \alpha_t + \beta_1 log\text{ total assets}_{bt-1} + \beta_2 log\text{ total assets}_{bt-1} \times \\ \text{D.Public} + \beta_3\text{D.CEO relevant education}_{bt} + \\ \beta_4\text{CEO meetings attended}_{bt} + \beta_5\text{CEO age}_{bt} + \epsilon_{bt} \end{aligned} \tag{6}$$

where log CEO pay is alternatively total remuneration, share of the salary component, and share of the variable pay component. I then use the residuals $\widehat{\epsilon}_{bt}$ as the regressors in equation 8 below. I also include squares of the residuals to see if there is any evidence of a quadratic relationship between remuneration components and bank risk.

$$\begin{aligned} log\ z - \text{score}_{b,t} = \alpha_b + \gamma_t + \boldsymbol{\phi_1}\widehat{\epsilon}_{b,t-1} + \boldsymbol{\phi_2}\widehat{\epsilon}_{b,t-1} \times D.public_b \\ + \boldsymbol{\phi_3}\widehat{\epsilon}^2_{b,t-1} + \boldsymbol{\phi_4}\widehat{\epsilon}^2_{b,t-1} \times D.public_b + \beta X_{b,t-1} + \epsilon_{b,t} \end{aligned} \tag{7}$$

[49] This result is sensitive to clustering, as it becomes significant at the 1% level when standard errors are clustered by time.

The results are presented in table 11. I find that total variable component residuals continue to be significantly (at 1%) positively related to z-scores private sector banks. On the other hand it is significantly (also at 1%) negatively related to public sector z-scores. The coefficients on the residual squares are significant and in the same direction, indicating a monotonic relationship. These results hold when I include CEO tenure in the model; or when I add some of the explanatory variables from the main model into equation 7, such as net interest margins or NNPAs.

7 Conclusions and discussion on next steps

In this paper, I study corporate governance in Indian banks along three dimensions. First, I provide detailed descriptions of how state bank boards in India look like, and along what dimensions they differ from private bank boards. This gives a sense of how constrained state banks truly are. Second, I study how shareholders respond to appointments and cessations of CEOs and other independent directors for the two sets of banks. Finally, I provide early evidence that for public banks, increasing CEO tenures relative to the board, and decreasing their variable pay component in total pay (or designing it differently) can improve the safety of these banks. I also find that older CEOs with greater experience are linked to lesser risk.

The work on this paper is relatively preliminary. However, it provides a base upon which to develop a longer-term research agenda, because this area is relatively under-researched, timely, and has significant policy implications. There are several ways to improve upon the existing analysis. The model can be improved upon by adding more information on the risk-aversion of the CEO and of the board as a whole. One potential idea is to use the inflation rate experienced by the director during her formative years as a proxy for the same, with the hypothesis being that the higher the inflation rate experienced, the higher would be the director's conservativeness. This idea is along the lines of Bernile *et al.* (2017), who use early life disasters as a measure of CEO risk-aversion. They find that CEO's risk aversion is linked to the intensity of their early-life experiences: CEOs that experience severe disasters without significant negative consequences behave more aggressively, while those who suffer greater negative consequences are more conservative. The event study analysis can also be expanded. I am interested in particular in how shareholders of private banks respond when government or central bank appointees are hired to sit on their boards.

The main takeaway from the work so far is that public banks in India need to do better in order to incentivise bank boards to be effective monitors of emerging risks. The solution to this is not one dimensional, but rather a combination of several policies. For example, variable pay may not be associated with greater risk if it is designed in a different way, or if the tenure of the CEO is not restricted. A good starting point might be to heed the recommendations of the Bank Boards Bureau in India.

References

Adams R, Ferreira D (2007). "A theory of friendly boards." *The Journal of Finance*, **LXII**(1).

Adams R, Ferreira D (2009). "Women in the boardroom and their impact on governance and performance." *Journal of Financial Economics*, **94**(2), 291–309.

Adams R, Mehran H (2011). "Corporate performance, board structure, and their determinants in the banking industry." *Staff Report No. 330*, Federal Reserve Bank of New York.

Aebi V, Sabato G, Schmid M (2012). "Risk management, corporate governance, and bank performance in the financial crisis." *Journal of Banking and Finance*, **36**(12), 3213–3226.

Ahern K, Dittmar AK (2012). "The changing of the boards: The impact on firm valuation of mandated female board representation." *The Quarterly Journal of Economics*, **127**(1), 137–197.

Amihud Y, Lev B (1981). "Risk reduction as a managerial motive for conglomerate mergers." *Bell Journal of Economics*, **12**(2), 605–617.

Andries AM, Nistor Mutu S, Ongena S, Sprincean N (2019). "On Becoming an O-SII ("Other Systemically Important Institution")." *Working paper*, SSRN. URL https://ssrn.com/abstract=3019888.

Anginer D, Demirguc-Kunt A, Huizinga H, Ma K (2016). "Corporate governance and bank capitalization strategies." *Journal of Financial Intermediation*, **26**(2016), 1–27.

Bandyopadhyay T (2017). "A paper tiger called Banks Board Bureau." *LiveMint*, **27 Feb, 2017**. URL https://www.livemint.com/Opinion/xI2UOoynj6CMW9yQTkji4L/A-paper-tiger-called-Banks-Board-Bureau.html.

Berger AN, Kick T, Schaeck K (2014). "Executive board composition and bank risk taking." *Journal of Corporate Finance*, **28**(C), 48–65.

Bernile G, Bhagwat V, Rau R (2017). "What doesn't kill you will only make you more risk-loving: Early-life disasters and CEO behavior." *Journal of Finance*, **72**(1), 167–206.

Block D, Gerstner AM (2016). "One-tier vs. two-tier board structure: A comparison between the united States and Germany." *Comparative Corporate Governance and Financial Regulation Paper I*, University of Pennsylvania Law School.

Bonnier KA, Bruner RF (1989). "An analysis of stock price reaction to management change in distressed firms." *Journal of Accounting and Economics*, **11**(1), 95–106.

Chen CR, Steiner TL, Whyte AM (2006). "Does stock option-based executive compensation induce risk-taking? An analysis of the banking industry." *Journal of Banking & Finance*, **30**(3), 915–945.

Cole S (2009). "Fixing market failures or fixing elections? Agricultural credit in India." *American Economic Journal: Applied Economics*, **1**(1), 219–50.

Coles J, Daniel ND, Naveen L (2006). "Managerial incentives and risk-taking." *Journal of Financial Economics*, **79**(2), 431–468.

de Andres P, Vallelado E (2008). "Corporate governance in banking: The role of the board of directors." *Journal of Banking and Finance*, **32**(2008), 2570–2580.

de Haan J, Vlahu R (2016). "Corporate governance of banks: A survey." *Journal of Economic Surveys*, **30**(2), 228–277.

Erkens DH, Hung M, Matos P (2012). "Corporate governance in the 2007-2008 financial crisis: Evidence from financial institutions worldwide." *Journal of Corporate Finance*, **18**(2), 389–411.

Faccio M, Marchica MT, Mura R (2016). "CEO gender, corporate risk-taking, and the efficiency of capital allocation." *Journal of Corporate Finance*, **39**(C), 193–209.

Fahlenbrach R, Stulz R (2011). "Bank CEO incentives and the credit crisis." *Journal of Financial Economics*, **99**(1), 11–26.

Falato A, Li D, Milbourn T (2015). "Which skills matter in the market for CEOs? Evidence from pay for CEO credentials." *Management Science*, **61**(12), 2845–2869.

Faleye O, Krishnan K (2017). "Risky lending: Does bank corporate governance matter?" *Journal of Banking and Finance*, **83**(2017), 57–69.

Ferrari G, Ferraro V, Profeta P, Pronzato C (2017). "Gender quotas: Challenging the boards, performance, and the stock market." *CHILD Working Papers Series 43*, Centre for Household, Income, Labour and Demographic Economics (CHILD) - CCA.

Gürkaynak R, Wright JH (2013). "Identification and inference using event studies." *Manchester School*, **81**, 48–65.

Hambrick DC, Mason PA (1984). "Upper echelons: The organization as a reflection of its top managers." *The Academy of Management Review*, **9**(2), 193–206.

Hau H, Thum M (2009). "Subprime crisis and board (in-) competence: Private versus public banks in Germany." *Economic Policy*, **24**(60), 701–752.

Hertzberg A, Liberti J, Paravisini D (2010). "Information and incentives inside the firm: Evidence from loan officer rotation." *Journal of Finance*, **65**(3), 795–828.

Huson MR, Malatesta P, Parrino R (2004). "Managerial succession and firm performance." *Journal of Financial Economics*, **74**(2), 237–275.

IiAS (2018). "Frauds in the Indian banking sector." *Technical report*, Institutional Investor Advisory Services. URL https://www.iiasadvisory.com/single-post/2018/03/29/Frauds-in-the-Indian-Banking-Sector.

IMF (2018). "India: Financial sector assessment program." *IMF Country Report No. 18/4*, International Monetary Fund. *Detailed assessment of observance of the Basel core principles for effective banking supervision.*

Jensen MC (1993). "The modern Industrial Revolution, exit, and the failure of internal control systems." *The Journal of Finance*, **XLVIII**(3).

Kim Y (1996). "Long-term firm performance and chief executive turnover: An empirical study of the dynamics." *Journal of Law, Economics, and Organization*, **12**(2), 480–496.

King T, Srivastav A, Williams J (2016). "What's in an education? Implications of CEO education for bank performance." *Journal of Corporate Finance*, **37**(C), 287–308.

Laeven L, Levine R (2009). "Bank governance, regulation and risk taking." *Journal of Financial Economics*, **93**(2), 259–275.

Lee PM, James EH (2007). "She-E-Os: Gender effects and investor reactions to the announcements of top executive announcements." *Strategic Management Journal*, **28**, 227–241.

MacKinlay A (1997). "Event studies in economics and finance." *Journal of Economic Literature*, **35**, 13–39.

Martin AD, Nishikawa T, Williams MA (2009). "CEO Gender: Effects on Valuation and Risk." *Quarterly Journal of Finance and Accounting*, **48**(3), 23–40.

MCA (2013). "The Companies Act, 2013." *Ministry of Corporate Affairs, Government of India.* URL http://www.mca.gov.in/Ministry/pdf/CompaniesAct2013.pdf.

Ministry of Finance (2017). "Economic Survey of India, 2016-17."

MoF (1955). "The Banking Companies (Second Amendment) Act, 1960." *Ministry of Finance, Government of India.*

MoF (1970). "The Banking Companies (Acquisition and Transfer of Undertakings) Act, 1970." *Ministry of Finance, Government of India.*

MoF (1980). "The Banking Companies (Acquisition and Transfer of Undertakings) Act, 1980." *Ministry of Finance, Government of India.*

MoF (2015). "Indradhanush plan for revamp of public sector banks." *Notification*, Department of Financial Services, Ministry of Finance, Government of India. URL http://financialservices.gov.in/sites/default/files/PressnoteIndardhanush_1.pdf.

Murphy K (1999). "Executive compensation." In O Ashenfelter, D Card (eds.), "Handbook of Labor Economics," volume 3, Part B, chapter 38, pp. 2485–2563. Elsevier, 1 edition.

Nayak Committee Report (2014). "Report of the committee to review governance of boards of banks in India." *Technical report*, Reserve Bank of India. URL https://rbidocs.rbi.org.in/rdocs/PublicationReport/Pdfs/BCF090514FR.pdf.

Nguyen BD, Nielsen KM (2010). "The value of independent directors: Evidence from sudden deaths." *Journal of Financial Economics*, **98**(3), 550–567.

Nguyen DD, Hagendorff J, Eshraghi A (2015). "Which executive characteristics create value in banking? Evidence from appointment announcements." *Corporate Governance: An International Review*, **2**(23), 112–128.

Nguyen DD, Hagendorff J, Eshraghi A (2016). "Can bank boards prevent misconduct?" *Review of Finance*, **20**(1), 1–36.

Ongena S, Savaser T, Şişli-Ciamarra E (2018). "CEO Incentives and Bank Risk over the Business Cycle." Unpublished manuscript; under review, URL https://bit.ly/2JZ3Xoc.

Pessarossi P, Weill L (2013). "Does CEO turnover matter in China? Evidence from the stock market." *Journal of Economics and Business*, **70**(C), 27–42.

Pourciau S (1993). "Earnings management and nonroutine executive changes." *Journal of Accounting and Economics*, **16**(1-3), 317–336.

RBI (1949). "The Banking Regulation Act, 1949." *Reserve Bank of India.*

RBI (2018). "Report on trend and progress of banking in India, 2017-18." *Technical report*, Reserve Bank of India. URL https://bit.ly/2WTDBde.

Rosenstein S, Wyatt JG (1990). "Outside directors, board independence, and shareholder wealth." *Journal of Financial Economics*, **26**(2), 175–191.

Sarkar A, Subramanian K, Tantri P (2019). "Effects of CEO turnover in banks: Evidence using exogenous turnovers in Indian banks." *Journal of Financial and Quantitative Analysis*, **54**(1), 183–214.

Sarkar J, Sarkar S (2018). "Bank ownership, board characteristics and performance: Evidence from commercial banks in India." *International Journal of Financial Studies*, **6**(1), 1–30.

SEBI (2000). "Addition of clause 49 to the listing agreement." *Circular*, Securities and Exchange Board of India. Based on the recommendations of the Committee on Corporate Governance headed by Shri Kumar Mangalam Birla (member, SEBI Board) to raise the standard of corporate governance of listed companies.

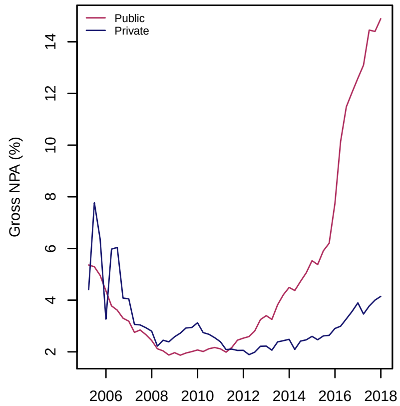
Walker SD (2009). "A review of corporate governance in UK banks and other financial industry entities: Final recommendations." *Technical report*, HM Treasury. Review report commissioned by U.K. government to recommend measures to improve board level governance. Final recommendations submitted on 26.11.2009.

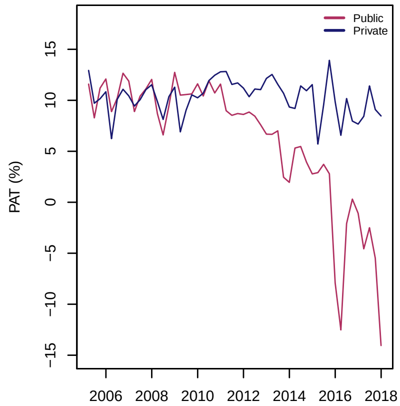
Wang G, Holmes Jr RM, Oh IS, Zhu W (2016). "Do CEOs matter to firm strategic actions and firm performance? A meta-analytic investigation based on upper-echelons theory." *Personnel Psychology*, **69**(4), 775–862.

Wolfers J (2006). "Diagnosing Discrimination: Stock Returns and Ceo Gender." *Journal of the European Economic Association*, **4**(2/3), 531–541.

Figures and tables

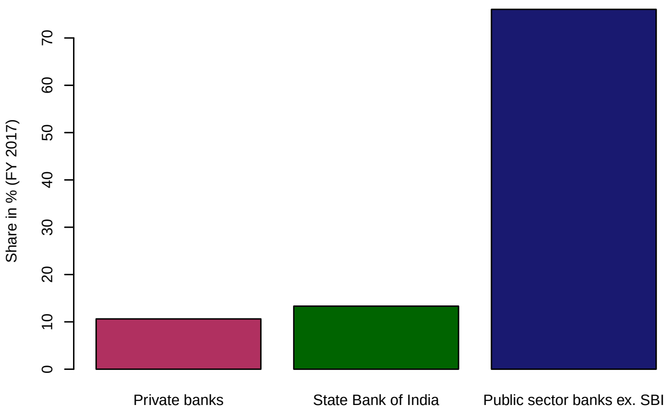
Figure 1: Performance of Indian banks, 2006–2018





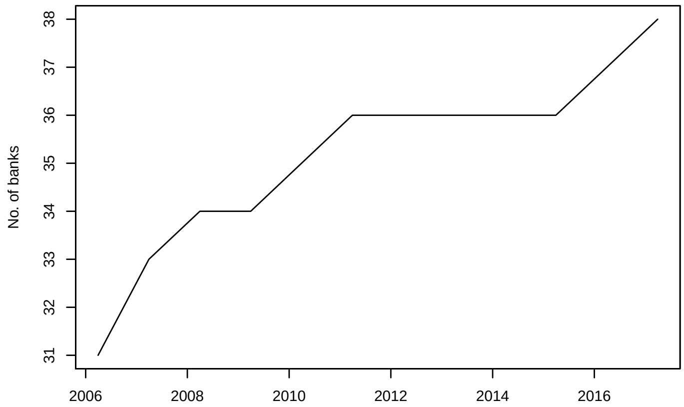
Note: This graph shows the gross non performing assets as share of total advances in the left panel, and the profits after tax in the right panel. The sample is all listed banks in India.

Figure 2: Share by type of bank in aggregate frauds, 2017



Note: This graph shows the share of private and public sector banks in total frauds totalling \$2.5bn in 2017. *State Bank of India*, a public bank, is the largest bank in India by total assets and one of the 100 largest banks in the world. The data is reproduced from IiAS (2018).

Figure 3: Number of banks per year



Note: This graph shows number of banks per year in the sample, between 2006–2017.

Table 1: Mergers and acquisitions in Indian banks, 2000–2017

Bank	Merger/ acquisition bank	Date
State Bank of Bikaner & Jaipur	SBI	01 Apr 2017
State Bank of Mysore	SBI	01 Apr 2017
State Bank of Travancore	SBI	01 Apr 2017
ING Vysya Bank	Kotak Mahindra Bank	01 Apr 2015
Bank of Rajasthan	ICICI Bank	12 Aug 2010
Centurion Bank of Punjab	HDFC Bank	23 May 2008
United Western Bank	IDBI Bank	12 Sep 2006
Bank of Punjab	Centurion Bank	29 Jun 2005
Global Trust Bank	Oriental Bank of Commerce	14 Aug 2004
Nedungadi Bank	Punjab National Bank	01 Feb 2003
Bank of Madura	ICICI Bank	10 Mar 2001
Times Bank	HDFC Bank	26 Feb 2000

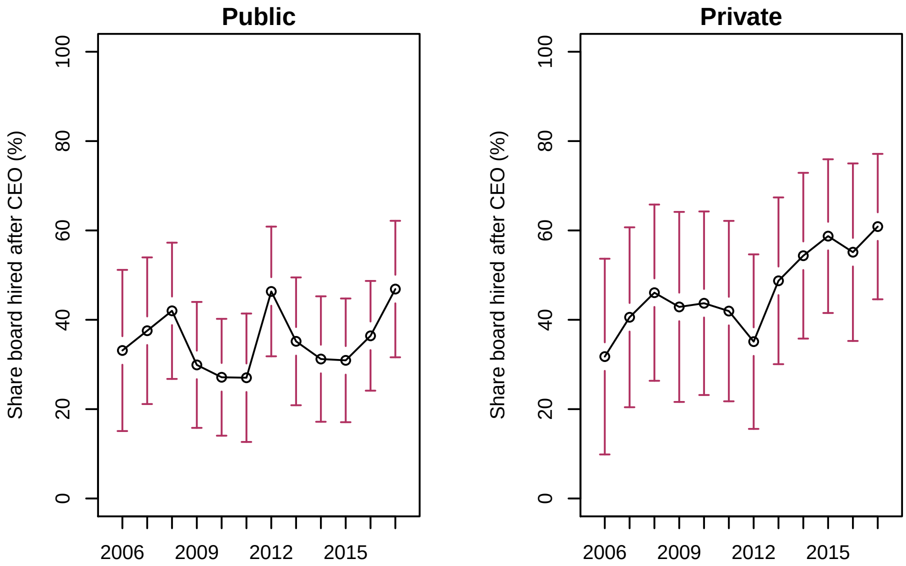
Note: The table shows mergers and acquisitions in the Indian banking sector. Only three overlap with the time period which I am studying, and I will use data that accounts for these mergers. There have been no bank failures over the sample period.

Table 2: Characteristics of the median Indian bank board, 2006–2017

Variable	**Range**	**State**	**Private**	$\overline{x}_s - \overline{x}_p$ **significant?** ($\alpha = 0.05$)
		Panel A: Median director		
Age				
Active director	28 - 82 yrs	58 yrs	61 yrs	Yes
Inactive director	30 - 82 yrs	58 yrs	63 yrs	Yes
Length of tenure	0 - 25 yrs	3.00 yrs	6.00 yrs	Yes
CEO tenure	1 - 10 yrs	2.00 yrs	5.00 yrs	Yes
Chairperson tenure	1 - 19 yrs	2.00 yrs	7.00 yrs	Yes
Board chair in other firms	0 - 26	2.50	1.00	Yes
Director in other firms	0 - 22	2.00	3.00	Yes
No. of meetings attended	2 - 26	9.83	7.8	Yes
No. of sub-committee positions held	0 - 12	4	2	Yes
Total remuneration (real, 2015 INR)	$1168-$315,000	$15,010	$51,000	Yes
		Panel B: Median bank		
Size of board	1 - 19	11	10	Yes
Share hired after CEO	0 - 93%	30%	55%	Yes
Share of women	0 - 27%	8.33%	2.77%	Yes
Share of independent directors	0 - 100%	0%	57.14%	Yes
Share of executive directors	0 - 83%	23.08%	12.5%	Yes
Dirs. on audit sub-committee	0 - 15	8	5	Yes
Dirs. on risk sub-committee	0 - 22	7	5	Yes
Dirs. on NPA sub-committee	0 - 10	0	0	Yes
Share with relevant education	0-100%	56%	70%	Yes

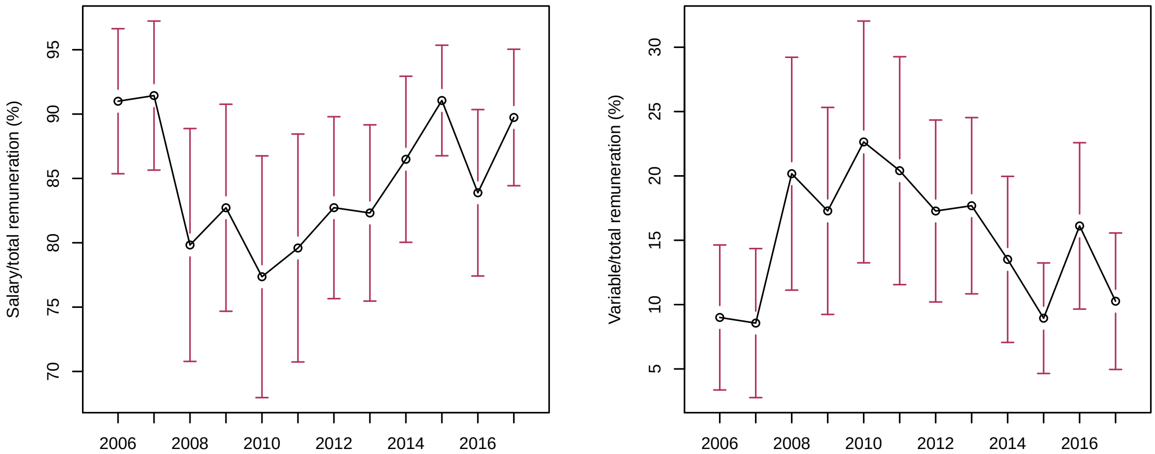
Note: The table reports various corporate governance characteristics for the median director of a public and private sector bank. Panel A reports statistics on the median director; whilst panel B reports values for the median bank. In the final column, I report whether the *t*-statistic of a difference in the averages between public and private banks for the variable was significant at the 5% level or not. See section 4 for a description and discussion on the variables.

Figure 5: CEO "soft" power



Note: This figure shows the CEO's "soft" power over the board for public (left panel) and private sector (right panel) banks. It is measured as the share of current board members that were appointed after the serving CEO.

Figure 4: CEO & chairperson pay: Salary vs. variable pay, as share of total remuneration



Note: This graph shows the components of total remuneration for CEO/CMD and chairperson, the two most senior positions on the board. The left hand panel shows the cash or salary to total remuneration and the right panel shows share of variable pay to total remuneration for all CEOs in the sample, over the period 2006–2017. The two do not necessarily add up to 100% due to sample differences.

Table 3: Reasons why directors leave bank boards, 2006–2017

Reason for leaving	No. of directors
Retired/Superannuated/Term Expired	577
Nomination Ended	145
Resigned (No reason stated)	99
Suspended/Terminated/Superceded	27
Demise	9
Nomination Withdrawn	7
Resigned (To Join other Board)	7
Vacation of Office	6
Resigned (Due to Personal Reason)	4
Resigned (Due to pre-occupation)	4
Resignation of Original Director	1
Resigned (Due to change in global role)	1
Resigned (Due to old age)	1
Total	888

Note: The table outlines the reasons why directors leave a bank's board. The age of retirement for public sector banks is 60 years, whilst it is not defined for private sector banks.

Table 4: Reasons why CEOs leave bank boards, 2006–2017

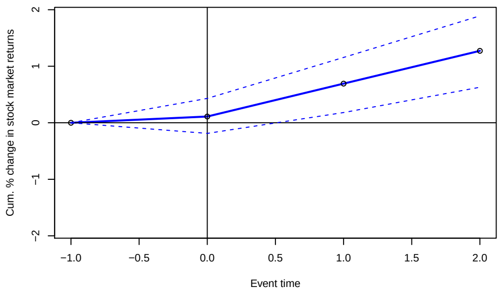
Reason for leaving	No. of CEOs
Retired/Superannuated/Term Expired	64
No information	16
Resigned (No reason stated)	6
Nomination Ended	2
Resigned (To Join other Board)	2
Suspended/Terminated/Superceded	2
Total	92

Note: The table shows the reasons why chief executive officers (CEOs) exit a bank. There are 120 unique CEOs in the sample, of which 36 are active at the end of the sample in 2017. There are 50 CEOs in the dataset who switch between banks in the sample (for example, as an executive director of one bank, and then CEO of another), of which atleast six that are moved to other banks as CEOs. This is why the total number of events in this table does not add up to 84 (the total inactive CEOs in end-2017).

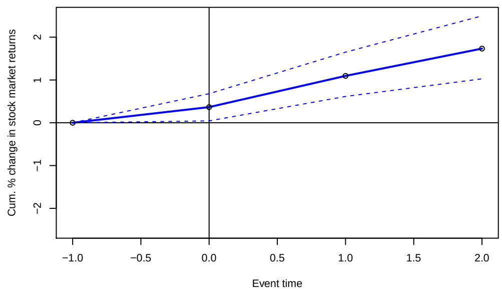
Table 5: Differences between public and private banks, 2006-2017

Variable	Mean: Public	Mean: Private	CI of difference	*p*-value
RWA Basel III (Rs.mn)	2433126.67	1293314.41	629739.29,1649885.23	0.00
Total loans (Rs.mn)	1980832.20	988942.42	777358.55,1206421.01	0.00
Total priority sector loans (Rs.mn)	501950.98	155386.61	283366.61,409762.13	0.00
Total income (Rs.mn)	56520.58	27060.59	24105.09,34814.89	0.00
Non-interest income (Rs.mn)	19354.74	9985.22	7332.78,11406.25	0.00
Gross NPA (%)	4.92	2.86	1.76,2.36	0.00
NIM (%)	2.60	3.32	-0.8,-0.63	0.00
ROA (%)	0.49	1.24	-0.9,-0.6	0.00
PAT to income (%)	5.97	10.27	-5.09,-3.51	0.00
PB (%)	0.95	2.21	-1.38,-1.14	0.00

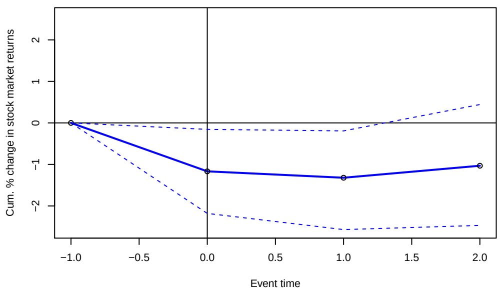
Figure 6: Cumulative abnormal percent change in stock market returns: CEO turnover



(a) All banks (175 events)



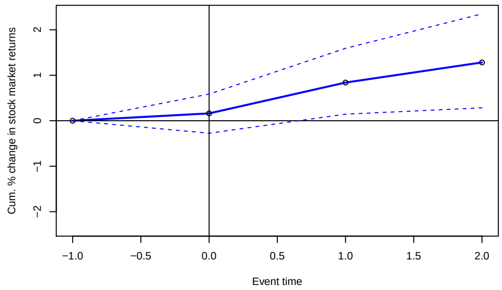
(b) Public banks (144 events)



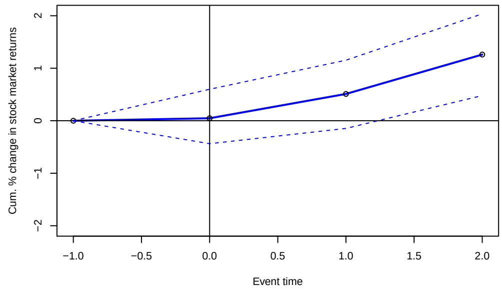
(c) Private banks (31 events)

Note: This graph shows the cumulative abnormal percent change in stock market returns for CEO turnover (both appointments and resignations) in (a) all banks (b) public sector banks and (c) private sector banks. The estimation window is 91 to 11 days before the event, and the cumulative abnormal returns are calculated over a window of three days, i.e. $(-1, +2)$, using a standard market model. The market index is *Nifty 50*, the Indian National Stock Exchange's benchmark broad based stock market index for the Indian equity market.

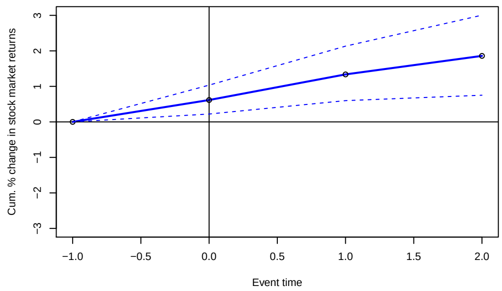
Figure 7: CEO appointment and resignation events



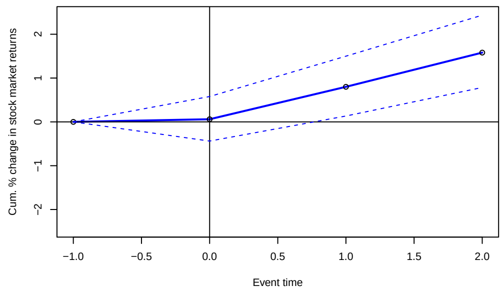
A1. All banks, CEO appointments (99 events)



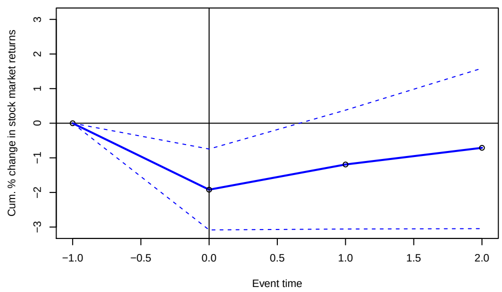
A2. All banks, CEO resignations (76 events)



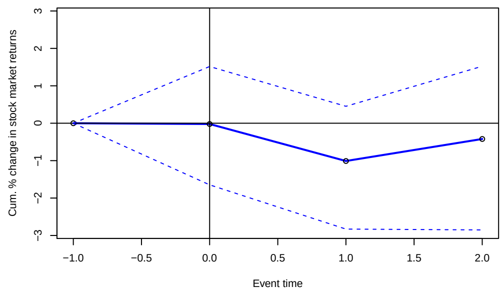
B1. Public banks, CEO appointments (80 events)



B2. Public banks, CEO resignations (64 events)



C1. Private banks, CEO appointments (19 events)



C2. Private banks, CEO resignations (12 events)

Note: This graph shows the cumulative abnormal percent change in stock market returns for CEO appointments and resignations separately in (A) all banks (B) public sector banks and (C) private sector banks. The estimation window is 91 to 11 days before the event, and the cumulative abnormal returns are calculated over a window of three days, i.e. $(-1, +2)$, using a standard market model. The market index is *Nifty 50*, the Indian National Stock Exchange's benchmark broad based stock market index for the Indian equity market.

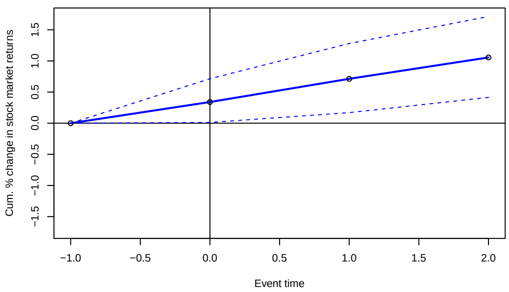
Table 6: Explaining public sector bank CEO turnover cumulative abnormal returns (CARs)

	Dependent variable:					
	Cum. Abnormal Returns (-1,+2)					
	(1)	(2)	(3)	(4)	(5)	(6)
Age at appointment	−0.015	−0.107	−0.113	−0.123	−0.135	−0.098
	(0.220)	(0.229)	(0.231)	(0.236)	(0.238)	(0.227)
Female	1.165**	1.362*	1.346*	1.352*	2.045**	1.526*
	(0.585)	(0.723)	(0.806)	(0.736)	(0.965)	(0.791)
log total assets	−0.919	−1.001*	−1.044	−1.052	−1.207*	−1.178*
	(0.565)	(0.525)	(0.646)	(0.647)	(0.661)	(0.644)
Return on assets		−4.134**	−4.231**	−4.321**	−4.495**	−4.206**
		(1.834)	(2.046)	(1.979)	(1.988)	(1.969)
Capital surplus (above 8%)		−0.056	−0.062	−0.079	−0.071	−0.047
		(0.224)	(0.240)	(0.243)	(0.227)	(0.236)
Net non performing assets		−1.318**	−1.325**	−1.367**	−1.351**	−1.353**
		(0.564)	(0.577)	(0.579)	(0.589)	(0.588)
Avg. board size			0.037	0.061	0.057	0.052
			(0.215)	(0.218)	(0.214)	(0.214)
Avg. number of meetings			0.103	0.091	0.125	0.109
			(0.138)	(0.137)	(0.136)	(0.140)
D.Masters			0.097			
			(0.751)			
D.PhD				3.143		
				(3.014)		
D.relevant education					1.189	
					(0.796)	
D.MBA						0.882
						(1.197)
Constant	28.518*	41.828***	42.043**	42.919**	46.981**	44.585**
	(14.565)	(14.071)	(17.625)	(16.975)	(18.080)	(18.182)
Observations	145	145	145	145	145	145
R^2	0.033	0.122	0.125	0.131	0.138	0.129
Adjusted R^2	0.012	0.083	0.066	0.073	0.080	0.071

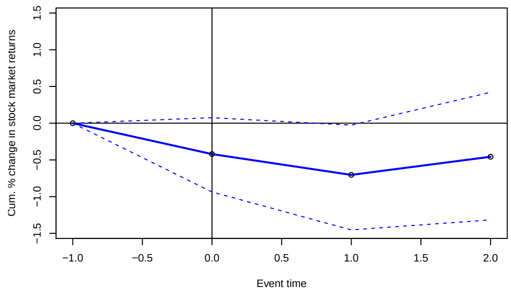
Robust standard errors in parentheses, clustered at director level..
*p<0.1; **p<0.05; ***p<0.01

Note: The table runs a cross-sectional regression of cumulative abnormal returns over a 3-day window around CEO turnover (CAR$(-1,+2)$) on CEO and bank-specific variables. The events are 145 CEO turnover events in public sector banks, i.e. both appointments and resignations, for 77 unique CEOs. *Age* is the age of the CEO at appointment. *Female* is a dummy variable when the CEO is female (4 CEOs, 6 events). There are four education variables, which are all dummies: *masters* if the CEO has a masters degree (41 CEOs), *PhD* if they have a doctorate (1 CEO), *MBA* if they have an MBA or a management degree (14 CEOs), *relevant* if they have a "relevant" qualification (52 CEOs). I consider any degree related to chartered accountancy, commerce, economics, management, MBA, or membership of the Indian Institute of Banking and Finance, as "relevant". The bank level controls are: *log* total assets, return on assets, capital as share of total assets less the Basel I 8% minimum, net non performing assets as share of total loans, average board size, and average number of meetings.

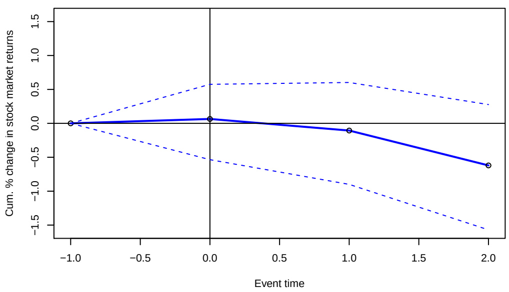
Figure 8: Independent director appointment and resignation events



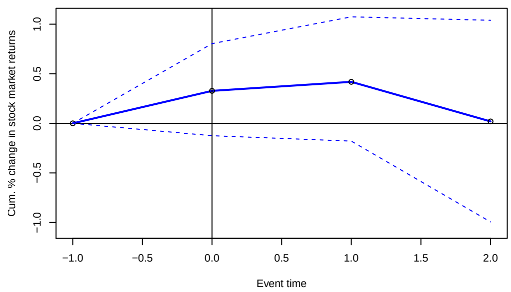
A1. Public banks, Independent dir. appointments



A2. Public banks, Independent dir. resignations



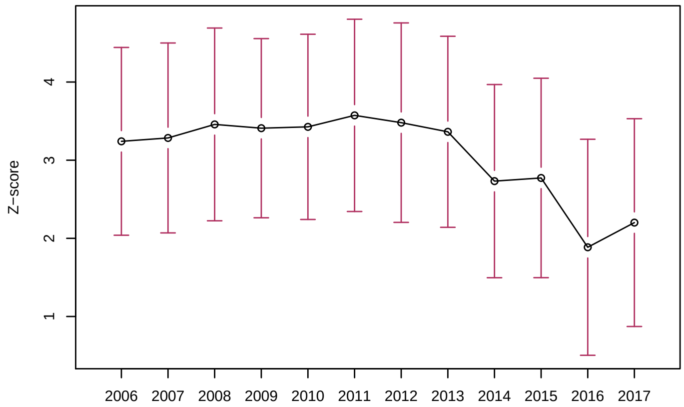
B1. Private banks, Independent dir. appointments



B2. Private banks, Independent dir. resignations

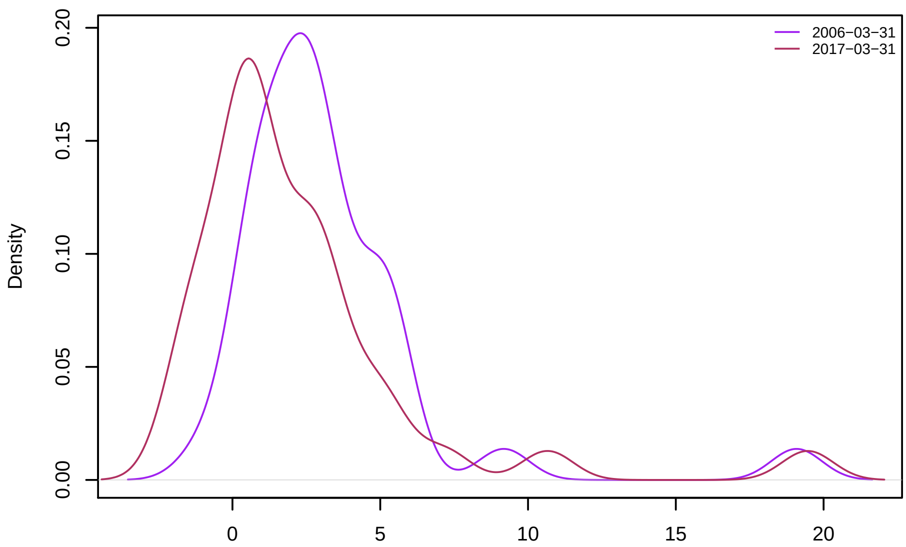
Note: This graph shows the cumulative abnormal percent change in stock market returns for independent director appointments and resignations separately in (A) public sector banks and (B) private sector banks. The estimation window is 91 to 11 days before the event, and the cumulative abnormal returns are calculated over a window of three days, i.e. $(-1, +2)$, using a standard market model. The market index is *Nifty 50*, the Indian National Stock Exchange's benchmark broad based stock market index for the Indian equity market.

Figure 9: Riskiness of Indian banks



Note: This graph shows the average z-score for the Indian banking sector, with 95% confidence bands.

Figure 10: Distribution of z-score



Note: The figure shows the z-score for the Indian banking sector at the end of the year in 2006 and 2017. z-score is defined as $\frac{\text{ROA}_{bt}+(\text{K/TA})_{bt}}{\sigma_{bt}^{ROA}}$, for bank b and time t. It measures the distance from insolvency for the bank and indicates the number of standard deviations that a bank's return on assets has to drop below its expected value before equity is depleted.

Table 7: Descriptive statistics

Statistic	N	Mean	St. Dev.	Pctl(25)	Median	Pctl(75)	Max
z-score	428	3.067	3.658	1.210	2.198	4.003	21.916
Market-to-book	431	1.486	1.305	0.745	1.141	1.631	11.369
Total assets ($\times 10^{12}$)	428	1.982	2.862	0.475	1.118	2.357	27.116
Net NPA/Total loans	428	1.965	2.202	0.640	1.185	2.322	13.990
Loan growth	428	0.167	0.124	0.099	0.170	0.236	0.590
Loans/Total assets	428	60.858	4.190	58.625	61.344	63.674	69.617
Credit to agriculture/Total assets	368	12.576	4.543	10.088	13.143	15.596	25.408
Non-interest income/Total income	427	35.351	5.827	31.245	34.255	39.012	57.131
Net interest margin	392	2.905	0.844	2.395	2.900	3.320	6.300
Non-residential financial institutional investors	428	15.076	16.671	1.100	10.300	21.025	73.400
Total meetings of the risk committee	409	4.352	1.405	4.000	4.000	5.000	11.000
Members of risk committee/Total board size	433	61.083	37.537	41.667	60.000	81.818	300.000
Avg. tenure of board	433	4.117	2.734	2.6	3	5.2	22
Avg. tenure of CEO	356	3.737	2.476	2.000	3.000	5.000	10.000
Avg. CEO tenure to board tenure	356	1.165	0.972	0.714	1.000	1.369	10.000
Inactive CEO tenure to active CEO age	332	0.063	0.041	0.034	0.051	0.083	0.192
Inactive board tenure to active board age	433	0.070	0.047	0.045	0.052	0.086	0.412
CEO salary/total remuneration	433	87.131	18.314	78.300	96.800	100.000	100.000
CEO total variable compensation/total remuneration	433	12.869	18.314	0	3.200	21.700	91.000
CEO total remuneration ($\times 10^6$, real 2015 INR)	433	8.90	15.20	1.50	2.50	6.60	94.70
CEO salary ($\times 10^6$, real 2015 INR)	433	6.74	10.71	1.29	1.97	6.25	61.63
CEO variable comp (real 2015 INR)	433	2.210	6.108	0	0.000	0.910	38.445

Table 8: Tenure and board meetings I

	Dependent variable:					
	log Z-score					
	(1)	(2)	(3)	(4)	(5)	(6)
log avg. CEO tenure$_{it-1}$	−0.066	0.251	−0.192	−0.008	−0.489**	−0.187
	(0.201)	(0.218)	(0.186)	(0.181)	(0.229)	(0.277)
log avg. CEO tenure$_{it-1}$ × *public*	0.330	−0.050	0.439**	0.151	0.677***	0.266
	(0.214)	(0.227)	(0.204)	(0.209)	(0.251)	(0.292)
log avg. board tenure$_{it-1}$	0.434**	0.493*	0.572**	0.763**	0.685***	0.745**
	(0.200)	(0.265)	(0.232)	(0.321)	(0.244)	(0.323)
log avg. board tenure$_{it-1}$× public	−0.839***	−1.027***	−0.923**	−1.269***	−1.028***	−1.216***
	(0.271)	(0.322)	(0.370)	(0.434)	(0.365)	(0.396)
Avg. board age$_{it-1}$					0.007	0.002
					(0.015)	(0.013)
Avg. CEO age$_{it-1}$					0.027*	0.034**
					(0.014)	(0.015)
log total assets$_{it-1}$	0.021	0.088	0.013	0.018	−0.040	−0.008
	(0.057)	(0.064)	(0.060)	(0.078)	(0.083)	(0.091)
NNPA/total assets$_{it-1}$	−0.070***	−0.063**	−0.057***	−0.067***	−0.058**	−0.046
	(0.019)	(0.028)	(0.018)	(0.024)	(0.024)	(0.032)
Non-interest income$_{it-1}$			0.011	0.020**	0.006	0.013**
			(0.007)	(0.009)	(0.006)	(0.007)
Net interest margin$_{it-1}$			0.214***	0.155**	0.195***	0.113*
			(0.065)	(0.076)	(0.051)	(0.058)
Loans/total assets$_{it-1}$					−0.015	−0.004
					(0.013)	(0.011)
Credit to sensitive sector$_{it-1}$					−0.004	−0.0004
					(0.006)	(0.006)
Non-resident FII$_{it-1}$					0.010	0.011
					(0.006)	(0.007)
Meetings of risk committee$_{it-1}$					0.010	−0.007
					(0.032)	(0.030)
Observations	322	322	297	297	263	263
R^2	0.304	0.112	0.392	0.211	0.435	0.298
Adjusted R^2	0.205	0.000	0.295	0.046	0.314	0.107
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Phase	Full	Phase	Full	Phase	Full

Robust standard errors in parentheses
*p<0.1; **p<0.05; ***p<0.01

Note: The dependent is $log\ z - score$, defined as $\frac{\text{ROA}_{bt}+(\text{K/TA})_{bt}}{\sigma_{bt}^{ROA}}$, for bank b at time t. The main variables of interest are: *board tenure* is the tenure of all inactive directors except the CEO; *CEO tenure* is the tenure of inactive CEO for the bank; and *board age* and *CEO age* are the ages of the active board members and CEO respectively. The control variables are: *log* total assets, net non performing assets to total loans, non-interest income, net interest margin, loans to assets, credit to sensitive sector (defined as lending to real estate, capital markets, and commodities), ownership by non-resident foreign institutional investors, meetings of the risk committee, share of women on the (active) board, and share with relevant education (defined as any qualification in chartered accountancy, commerce, economics, management, MBA, or membership of the Indian Institute of Banking and Finance). *Public* is a dummy that takes value 1 if the bank is state-owned. All specifications contain bank fixed effects. For columns (1), (3), (5) I include only "phase" fixed effects, where the time period is divided into two: 2006 to 2011, and 2012 to 2017. Columns (2), (4) and (6) contain full time fixed effects. Standard errors are clustered at the bank level.

Table 9: Tenure and board meetings II

	Dependent variable:			
	log Z-score			
	(1)	(2)	(3)	(4)
log avg. CEO tenure$_{it-1}$	-0.489^{**}	-0.208		
	(0.237)	(0.282)		
log avg. CEO tenure$_{it-1}$ × *public*	0.690^{***}	0.282		
	(0.251)	(0.293)		
Avg. CEO age$_{it-1}$	0.027^{*}	0.036^{**}		
	(0.014)	(0.015)		
log avg. board tenure$_{it-1}$	0.680^{***}	0.745^{**}		
	(0.244)	(0.328)		
log avg. board tenure$_{it-1}$× public	-1.042^{***}	-1.269^{***}		
	(0.381)	(0.434)		
Avg. board age$_{it-1}$	0.005	0.001		
	(0.015)	(0.013)		
CEO tenure/ board tenure$_{it-1}$			-0.391^{***}	-0.343^{**}
			(0.125)	(0.148)
CEO tenure/ board tenure$_{it-1}$ × *public*			0.553^{***}	0.440^{***}
			(0.139)	(0.163)
CEO age/ board age$_{it-1}$			1.118	1.751^{**}
			(0.748)	(0.785)
CEO age/ board age$_{it-1}$ × *public*			-0.312	-1.056
			(1.205)	(1.178)
log total assets$_{it-1}$	-0.053	-0.038	-0.041	-0.058
	(0.103)	(0.107)	(0.102)	(0.120)
NNPA/Total assets$_{it-1}$	-0.058^{**}	-0.042	-0.060^{**}	-0.040
	(0.025)	(0.031)	(0.025)	(0.032)
Non-interest income$_{it-1}$	0.007	0.013^{**}	0.007	0.016^{**}
	(0.005)	(0.006)	(0.005)	(0.006)
Net interest margin$_{it-1}$	0.193^{***}	0.117^{*}	0.201^{***}	0.148^{**}
	(0.057)	(0.061)	(0.063)	(0.070)
Loans/total assets$_{it-1}$	-0.015	-0.004	-0.015	-0.005
	(0.012)	(0.011)	(0.012)	(0.012)
Credit to sensitive sector$_{it-1}$	-0.004	0.0002	-0.003	0.002
	(0.006)	(0.006)	(0.007)	(0.007)
Non-resident FII$_{it-1}$	0.010	0.011	0.012^{*}	0.013^{*}
	(0.006)	(0.007)	(0.007)	(0.007)
Meetings of risk committee$_{it-1}$	0.010	-0.007	0.016	-0.005
	(0.032)	(0.030)	(0.029)	(0.027)
Share female$_{it-1}$	0.005	0.007	0.005	0.007
	(0.005)	(0.005)	(0.005)	(0.006)
Share with relevant education$_{it-1}$	-0.064	0.043	-0.129	-0.031
	(0.201)	(0.185)	(0.196)	(0.186)
Observations	263	263	263	263
R^2	0.439	0.308	0.424	0.268
Adjusted R^2	0.313	0.111	0.301	0.069
Bank FE	Yes	Yes	Yes	Yes
Time FE	Phase	Full	Phase	Full

Robust standard errors in parentheses.
*p<0.1; **p<0.05; ***p<0.01

Note: The dependent is *log* $z - score$, defined as $\frac{\text{ROA}_{bt}+(\text{K/TA})_{bt}}{\sigma^{ROA}_{bt}}$, for bank b at time t. The main variables of interest are: *board tenure* is the tenure of all inactive directors except the CEO; *CEO tenure* is the tenure of inactive CEO for the bank; and *board age* and *CEO age* are the ages of the active board members and CEO respectively. *Relative age of CEO* is the age of the CEO divided by the average age of the board; similarly *relative tenure of CEO* is the tenure of the CEO divided by the average tenure of the board. The control variables are: log of total assets, net non performing assets to total loans, non-interest income, net interest margin, loans to assets, credit to sensitive sector (defined as lending to real estate, capital markets, and commodities), ownership by non-resident foreign institutional investors, meetings of the risk committee, share of women on the (active) board, and share with relevant education (defined as any qualification in chartered accountancy, commerce, economics, management, MBA, or membership of the Indian Institute of Banking and Finance). *Public* is a dummy that takes value 1 if the bank is state-owned. All specifications contain bank fixed effects. For columns (1) and (3) I include only "phase" fixed effects, where the time period is divided between 2006 and 2011, and 2012 to 2017. Columns (2) and (4) contain full time fixed effects. Standard errors are clustered at the bank level.

Table 10: CEO salary components I

	Dependent variable:			
	log Z-score			
	(1)	(2)	(3)	(4)
CEO salary to TR_{it-1}	-0.004^{**}			
	(0.002)			
CEO salary to $\text{TR}_{it-1} \times$ *public*	0.004^{*}			
	(0.003)			
CEO salary to TR_{it-3}		-0.009^{***}		
		(0.002)		
CEO salary to $\text{TR}_{it-3} \times$ *public*		0.009^{***}		
		(0.003)		
CEO TVC to TR_{it-1}			0.004^{**}	
			(0.002)	
CEO TVC to $\text{TR}_{it-1} \times$ *public*			-0.004^{*}	
			(0.003)	
CEO TVC to TR_{it-3}				0.009^{***}
				(0.002)
CEO TVC to $\text{TR}_{it-3} \times$ *public*				-0.009^{***}
				(0.003)
log total assets_{it-1}	0.123	0.118	0.121	0.127
	(0.088)	(0.085)	(0.087)	(0.086)
NNPA/total assets_{it-1}	-0.083^{***}	-0.084^{***}	-0.082^{***}	-0.084^{***}
	(0.020)	(0.020)	(0.019)	(0.020)
Non-interest income_{it-1}	0.015^{**}	0.016^{**}	0.014^{**}	0.015^{**}
	(0.006)	(0.006)	(0.007)	(0.006)
Net interest margin_{it-1}	0.119^{**}	0.103^{*}	0.120^{**}	0.107^{*}
	(0.057)	(0.056)	(0.052)	(0.055)
Loans/total assets_{it-1}	-0.006	-0.005	-0.007	-0.007
	(0.008)	(0.007)	(0.008)	(0.007)
Credit to sensitive sector_{it-1}	0.003	0.002	0.002	0.003
	(0.006)	(0.006)	(0.006)	(0.006)
Non-resident FII_{it-1}	0.005	0.005	0.005	0.005
	(0.004)	(0.005)	(0.004)	(0.004)
Meetings of risk committee_{it-1}	0.001	-0.001	0.004	0.002
	(0.022)	(0.022)	(0.021)	(0.023)
Avg. board size_{it-1}	-0.009	-0.012	-0.009	-0.008
	(0.015)	(0.014)	(0.014)	(0.014)
Share of female directors_{it-1}	0.003	0.004	0.002	0.003
	(0.005)	(0.005)	(0.005)	(0.005)
Share with relevant education_{it-1}	-0.321^{*}	-0.345^{**}	-0.313^{*}	-0.335^{*}
	(0.184)	(0.174)	(0.181)	(0.184)
Observations	345	343	345	343
R^2	0.198	0.224	0.198	0.224
Adjusted R^2	0.035	0.066	0.035	0.066
Bank FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes

Robust standard errors in parentheses.
*p<0.1; **p<0.05; ***p<0.01

Note: The dependent is *log* $z-$score, defined as before. The main variables of interest are components of the CEO's total remuneration (TR). *CEO salary* is the flat amount paid to the CEO and *CEO TVC* is the total variable component paid to the CEO (which may include bonuses, for example, but is different from bank to bank), defined as total remuneration less the salary component. Both are measured as share of total remuneration. The control variables are as before: *log* total assets, net non performing assets to total loans, non-interest income, net interest margin, loans to assets, credit to sensitive sector (defined as lending to real estate, capital markets, and commodities), ownership by non-resident foreign institutional investors, meetings of the risk committee, share of women on the (active) board, and share with relevant education (defined as any qualification in chartered accountancy, commerce, economics, management, MBA, or membership of the Indian Institute of Banking and Finance). Average (active) board size is also included as a control. *Public* is a dummy that takes value 1 if the bank is state-owned. Columns (2) and (4) lag the salary variables by three years, which is the median tenure length of a CEO in my dataset. All specifications contain bank and time fixed effects and standard errors are clustered at the bank level.

Table 11: CEO salary components II

	Dependent variable:		
	log Z-score		
	(1)	(2)	(3)
CEO TVC residuals$_{it-1}$	0.043***		
	(0.013)		
CEO TVC residuals$^2_{it-1}$	0.005***		
	(0.002)		
CEO TVC residuals$_{it-1}$ × *public*	−0.049***		
	(0.017)		
CEO TVC residuals$^2_{it-1}$ × *public*	−0.005**		
	(0.002)		
CEO salary residuals$_{it-1}$		0.129	
		(0.095)	
CEO salary residuals$^2_{it-1}$		0.049	
		(0.035)	
CEO salary residuals$_{it-1}$ × *public*		−0.084	
		(0.115)	
CEO salary residuals$^2_{it-1}$ × *public*		−0.074	
		(0.053)	
CEO TR residuals$_{it-1}$			0.224*
			(0.136)
CEO TR residuals$^2_{it-1}$			0.085*
			(0.047)
CEO TR residuals$_{it-1}$ × *public*			−0.198
			(0.155)
CEO TR residuals$^2_{it-1}$ × *public*			−0.093
			(0.060)
NNPA/total assets$_{it-1}$	−0.072***	−0.074***	−0.073***
	(0.022)	(0.024)	(0.024)
Non-interest income$_{it-1}$	0.011	0.014**	0.013*
	(0.007)	(0.007)	(0.006)
Net interest margin$_{it-1}$	0.142*	0.084	0.061
	(0.075)	(0.069)	(0.070)
Share with relevant education$_{it-1}$	−0.356*	−0.325	−0.350*
	(0.211)	(0.199)	(0.210)
All other controls	Yes	Yes	Yes
Observations	299	307	307
R^2	0.215	0.186	0.204
Adjusted R^2	0.029	0.000	0.021
Bank FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes

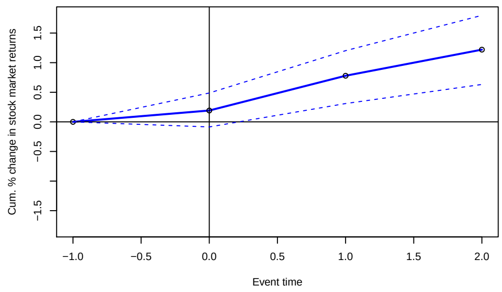
Robust standard errors in parentheses.
*p<0.1; **p<0.05; ***p<0.01

Note: The dependent is $log\ z-$score, defined as before. CEOs total remuneration, salary, and total variable pay (in real 2015 Indian Rupees) are regressed on total assets, dummy if the CEO has relevant education, CEO's meetings attended, and CEO's age (see equation 7). The drop in observations is because of incomplete information on meetings attended by CEOs. Residuals from the corresponding models are used as the main explanatory variables. The control variables are as before: *log* total assets, net non performing assets to total loans, non-interest income, net interest margin, loans to assets, credit to sensitive sector (defined as lending to real estate, capital markets, and commodities), ownership by non-resident foreign institutional investors, meetings of the risk committee, share of women on the (active) board, and share with relevant education (defined as any qualification in chartered accountancy, commerce, economics, management, MBA, or membership of the Indian Institute of Banking and Finance). Average (active) board size is also included as a control. *Public* is a dummy that takes value 1 if the bank is state-owned. All specifications contain all controls but only select few are reported. All columns contain bank and time fixed effects and standard errors are clustered at the bank level.

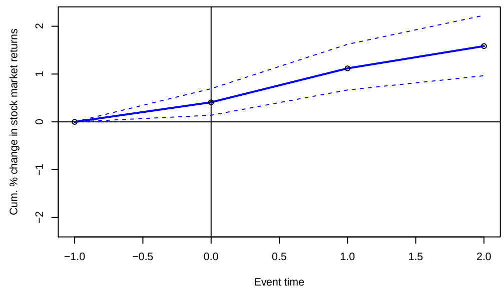
Appendices

A Additional figures

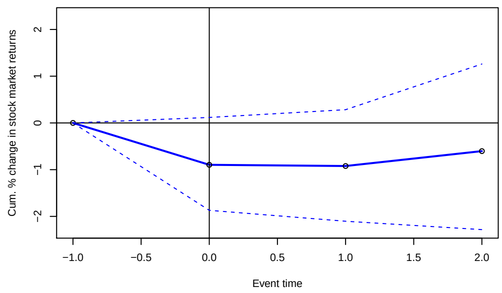
Figure A.1: Robustness: CEO turnover cumulative abnormal returns with alternate market index



(a) All banks (175 events)



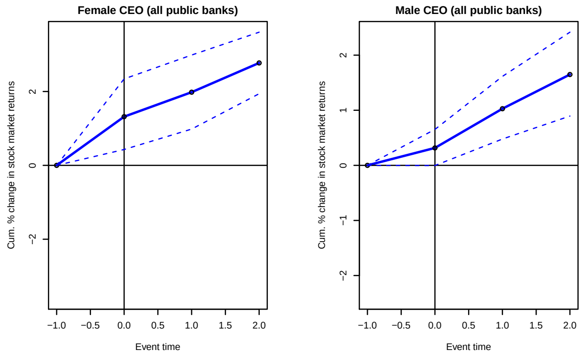
(b) Public banks (144 events)



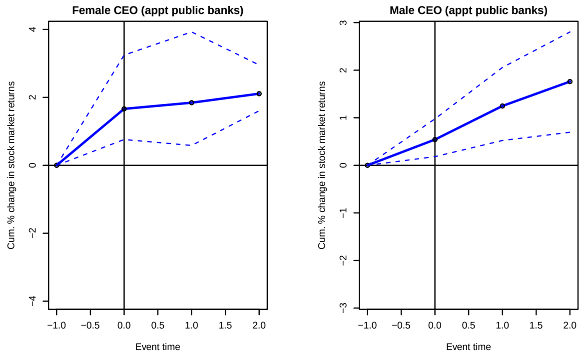
(c) Private banks (31 events)

Note: This figure is a robustness check on figure 6. It shows the cumulative abnormal percent change in stock market returns for CEO turnover (both appointments and resignations) in (a) all banks (b) public sector banks and (c) private sector banks. The estimation window is 91 to 11 days before the event, and the cumulative abnormal returns are calculated over a window of three days, i.e. $(-1, +2)$, using a standard market model. The market index is *Nifty Bank*, the Indian National Stock Exchange's benchmark for Indian banking stocks. It is comprised of 12 of the most liquid and large capitalised Indian banks.

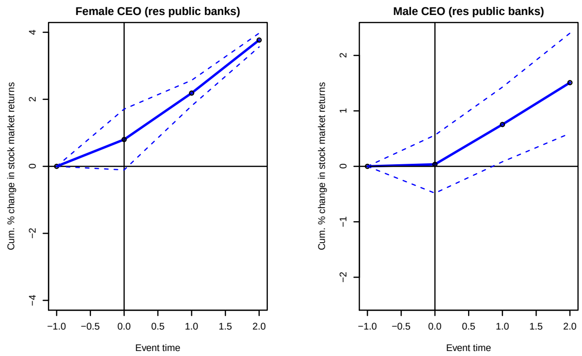
Figure A.2: Additional results: Public sector banks CEO turnover by gender



A. Pooling all events



B. Appointments



C. Leave

Note: This figure presents an additional event study in support of table 6. Events in left columns are defined as all female CEO related turnover in public banks, which gives us a total of 6 events (panel A), 4 appointments (panel B), and 2 resignations (panel C). Events in the right hand column are male CEO turnover, for a total of 141 events (panel A), of which there are 79 appointments (panel B) and 62 resignations (panel C).